



## Research paper

## Evaluating classical time-series models ARIMA, SARIMA, ARIMAX, and VAR for wave height forecasting

Ashraf Zaghwan<sup>a,\*</sup>, Hossein Enshaei<sup>a</sup>, Hamed Majidiyan<sup>a</sup>, Yousef Amer<sup>b</sup><sup>a</sup> College of Science and Engineering, Australian Maritime College, University of Tasmania, Launceston, Tasmania, 7250, Australia<sup>b</sup> Faculty of Science, Engineering and Technology, Adelaide University, Australia

## ARTICLE INFO

## Keywords:

Wave forecasting  
Time-series analysis  
ARIMA  
ARIMAX  
SARIMA  
VAR  
Ship routing  
Decision support system

## ABSTRACT

This study evaluates the suitability of classical time-series models for forecasting significant wave heights using real hydrometeorological observations, with the objective of improving data-driven operational decision-making in ship routing. Accurate short-term wave forecasts can enhance navigational safety, reduce fuel consumption, and minimize emissions. However, many existing Decision Support Systems (DSS) lack integrated forecasting capability to incorporate credible environmental datasets. To address this gap, this research develops a time series-based analytical framework employing four classical forecasting models: Autoregressive Integrated Moving Average (ARIMA), Seasonal ARIMA (SARIMA), ARIMA with Exogenous Variables (ARIMAX), and Vector Autoregressive (VAR). Models were trained and evaluated using observational data from the National Data Buoy Center station 41049 (South Bermuda) under both static and dynamic scenarios. The results indicate limited evidence of stable seasonality in the short-term wave record, and SARIMA showed weaker performance with higher percentage errors. ARIMAX provided comparatively better accuracy due to the influence of exogenous meteorological drivers, while overall findings support the null hypothesis that classical linear models have a restricted capability in capturing rapid wave fluctuations. These insights highlight the practical challenges of applying classical forecasting models to complex sea-state dynamics and underline the need for more advanced or hybrid approaches in DSS development for maritime operations.

## 1. Introduction

Reliable forecasting of sea states is fundamental to providing efficient ship operations. Wave height, period, and direction strongly influence vessel motion, fuel efficiency, voyage duration, and crew comfort. Yet, the high temporal and spatial variability of marine environments makes forecasting these parameters a continuing challenge. The accuracy of any predictive framework depends largely on the credibility of hydrometeorological data and on the analytical model's ability to capture temporal dependencies from historical observations. Several recent studies have demonstrated diverse data-driven approaches for improving navigation safety and operational forecasting in marine environments. Evidence shown in the literature that researchers have increasingly relied on historical data to extract meaningful navigation patterns and model complex vessel behaviours in open sea areas and near port approaches. In the near-port areas, traffic convergence creates high navigational risk. For instance, aligning the average

Automatic Identification System (AIS) trajectories, generating representative route centerlines, reflects real traffic density (Liu et al., 2023), (Rong et al., 2020), (Xu et al., 2025). Beyond route modeling, other studies have advanced predictive methods for vessel performance under varying environmental conditions. For instance, Guo et al. (2023) introduced the Domain-Adapted Feature Transfer (DAFT) framework. It combines AIS and met-ocean data to improve short-term vessel speed prediction by transforming inconsistent input variables, such as wind, wave, and swell, into domain-invariant features that enhance model generalization (Guo et al., 2023).

In parallel, a growing body of research has focused on evaluating and improving the credibility of hydro-meteorological datasets used in operational forecasting. Comparative assessments of global forecasting systems, such as those from: (1) National Centers for Environmental Prediction (NCEP), (2) Canadian Meteorological Centre (CMC), (3) European Centre for Medium-Range Weather Forecasts (ECMWF), and (4) German weather service Deutscher Wetterdienst DWD. Utilizing these

\* Corresponding author.

E-mail addresses: [ashraf.zaghwan@utas.edu.au](mailto:ashraf.zaghwan@utas.edu.au) (A. Zaghwan), [hossein.enshaei@utas.edu.au](mailto:hossein.enshaei@utas.edu.au) (H. Enshaei), [hamed.majidiyan@utas.edu.au](mailto:hamed.majidiyan@utas.edu.au) (H. Majidiyan), [yousef.amer@adelaide.edu.au](mailto:yousef.amer@adelaide.edu.au) (Y. Amer).<https://doi.org/10.1016/j.oceaneng.2025.124098>

Received 7 July 2025; Received in revised form 4 December 2025; Accepted 23 December 2025

Available online 8 January 2026

0029-8018/© 2026 The Authors. Published by Elsevier Ltd. This is an open access article under the CC BY license (<http://creativecommons.org/licenses/by/4.0/>).

models has demonstrated that ensemble-based forecasts provide more stable and reliable representations of metocean conditions, particularly when uncertainty quantification is crucial for informed decision-making (D'Agostini et al., 2024). Machine learning methods, including Random Forest (RF), Gradient Boosting Decision Trees (GBDT), and Support Vector Regression (SVR), have also been widely adopted for bias correction in hindcast datasets, significantly improving the accuracy of wind and wave predictions validated against buoy observation (Loake et al., 2022). Moreover, researchers have explored machine learning models for phase-resolved wave forecasting, addressing nonlinear and short-crested sea conditions by predicting down-wave surface elevations from multiple up-wave buoy measurements (Chen et al., 2023). Collectively, these studies, along with numerous other similar studies, highlight a clear evolution toward integrating historical data, statistical learning, and ensemble-based modeling to enhance navigational safety, improve forecast reliability, and support operational decisions in maritime engineering.

Time-series analysis offers a transparent, data-driven approach for interpreting ocean dynamics and forecasting short-term variability. Classical statistical models, such as the Autoregressive Integrated Moving Average (ARIMA) and its derivatives, have long been used to model stationary or near-stationary datasets, providing explainable relationships between parameters and manageable computational cost. Recent studies underline the continuing relevance of classical forecasting methods (Khairuddin et al., 2019; Wang et al., 2020; Katimon et al., 2018). For higher precision, Box and Jenkins (2015) recommended at least 40–50 data observations (Box et al., 2015). Mwenda et al. (2015) found that 120 past observations worked best for short-term forecasts, where 260 were more effective for medium-term horizons (Mwenda et al., 2015). Despite recent progress in hybrid and machine-learning approaches, classical models remain valuable for operational environments where interpretability, robustness, and real-time adaptability are essential.

In wave condition forecasting, short-term horizons often yield higher accuracy than long-term predictions due to the strong spatiotemporal variability of sea states driven by rapidly evolving wind-wave interactions. Capturing high-frequency fluctuations in parameters such as significant wave height ( $H_s$ ) and peak wave period ( $T_p$ ) over a few hours is more effective for navigation and ship routing, directly enhancing voyage safety and fuel efficiency compared to less reliable long-term forecasts. Despite evidence for short-term forecasting, few studies have compared multiple classical models using real buoy observations to evaluate their relative suitability for operational ship-routing applications. This study addresses this gap by assessing the forecasting performance of four classical models, ARIMA, Seasonal ARIMA (SARIMA), ARIMA with Exogenous Variables (ARIMAX), and Vector Autoregressive (VAR), using observational data from the National Data Buoy Centre (NDBC) station 41049 (South Bermuda) (please refer to the Data section) (NDBC - Station 41049 historical data, 2025). The analysis quantifies each model's ability to capture both short-term fluctuations and long-term trends in significant wave height, providing an analytical foundation for integrating reliable forecasts into maritime DSS frameworks (please refer to the DSS section). The primary objective is to assess the environmental characteristics of waves and develop a corresponding model-driven code (Nielsen and Stredulinsky, 2012). Aligning with the operational ship routing goals during sailing at sea. This study emphasizes “Operations” as the key focus on operational ship routing. Further elaboration on the choice of these models is introduced in the following research question, objectives, hypotheses (H0, H1, H2), and DSS model selection rationale.

### 1.1. Research question

How effective are classical time-series models (ARIMA, ARIMAX, SARIMA, and VAR) in forecasting sea-wave heights for integration into operational Decision Support Systems aimed at improving ship-routing

performance?

### 1.2. Objectives

1. Develop and validate a classical time-series forecasting framework using real hydrometeorological observations.
2. Compare the predictive capabilities of ARIMA, ARIMAX, SARIMA, and VAR models under static and dynamic forecasting conditions.
3. Evaluate the operational implications of forecast accuracy for ship routing and decision support.

### 1.3. Hypotheses

**H0.** Classical time-series models do not significantly capture wave-height variability and therefore do not enhance operational DSS performance.

**H1.** Classical models capture long-term trends in wave heights, partially improving DSS support for ship routing.

**H2.** Classical models capture both short-term fluctuations and long-term trends, providing reliable inputs for operational DSS integration.

### 1.4. Model selection rationale

Database research spanning from 1961 to 2025, using the keywords “Classical modeling” (CL), “Decision Support System”, and “Ship Routing”. 281 relevant journal articles were identified. Out of these, only 40 papers published between 1999 and 2024 were directly related to ship routing algorithms. Also, there was a notable rise in publications over the last decade. As shown in Table 1, the nominated papers focus on ship routing and decision-making using *classical models* (CL) algorithms. After reviewing these 25 articles, the Review indicated that the long-standing reliability and acceptance of CL algorithms in the field of ship routing research remain valid, utilizing several DSS tools. Despite CL models seeming to be desirable statistical approaches, only eight distinct CL algorithms were identified as practiced across twenty-six studies, as summarized in the references from (Dębski and Dreżewski, 2024; Du et al., 2021; Weintrit and Neumann, 2011; Nielsen and Dietz, 2020; Panigrahi et al., 2008; Harris et al., 2018; Hildeman et al., 2019, 2020, 2022; Roh, 2013a, 2013b; Ichsan et al., 2019; Son et al., 2020; Zhang et al., 2020; Yoon et al., 2023; Kunicka and Litwin, 2019; Furuichi and Otsuka, 2015; Cai and Wen, 2014; Sharif et al., 2024; Guo et al., 2019; Kulkarni et al., 2024; Sasmal et al., 2021; Cai et al., 2021; Panigrahi and Umesh, 2008; McCord et al., 1999; Hoorn and Knapp, 2019), in Table 1. This selection of algorithm options is utilized for ship routing simulations.

The consistent relevance of CL models across both newer and older studies highlights their enduring practicality (Ospina et al., 2023; Sahranavard and Naseri, 2022). In line with this, ARIMA, ARIMAX, SARIMA, and VAR were used in this study (Sahranavard and Naseri, 2022). Further elaboration on the choice of these models is introduced in the following section.

## 2. Decision Support System (DSS)

Amid Industry 4.0, IMO/IALA's e-Navigation, and GMTT 2030 framework, alongside the IMO's emerging MASS (Maritime Autonomous Surface Ships) CODE. These initiatives, in conjunction with addressing “10,000 Scientific Challenges,” are expected to derive intelligent ships equipped with decision support systems (DSSs) (Lee et al., 2022). The perspective of DSS is computer-based systems that enable users to leverage information, communication, documents, data, knowledge, and situational models to address complex problems, gain valuable insight, and make informed decisions (Liu et al., 2023). DSSs anticipate integrating sensor data, forecasts, and ship performance models to

**Table 1**  
Literature review of ship routing publications utilizing classical models.

| Ref.                        | Year | Algorithms                                                 | Method                                                            | DSS Suitability |
|-----------------------------|------|------------------------------------------------------------|-------------------------------------------------------------------|-----------------|
| Debski and Dreżewski (2024) | 2024 | EDA (Estimation of Distribution Algorithm)                 | Multi-objective optimization                                      | ✓               |
| Du et al. (2021)            | 2021 | Customize a Ship route estimation model                    | Mathematical modelling                                            | ✓               |
| Weinrit and Neumann (2011)  | 2011 | Powell's Method                                            | Bezier curves optimization                                        | ✓               |
| Nielsen and Dietz (2020)    | 2020 | ill-posed equations, stability in sea state estimation     | Bayesian Modeling                                                 | ✓               |
| Panigrahi et al. (2008)     | 2008 | Dijkstra, IRS-P4 satellite, 3G-WAM                         | Optimization technique                                            | ✓               |
| Harris et al. (2018)        | 2018 | Excel Microsoft                                            | Scenario modeling & packaging options                             | ✓               |
| Hildeman et al. (2022)      | 2022 | SPDE, Stepwise max. likelihood                             | Bivariate Gaussian Random Field                                   | ✓               |
| Roh (2013a)                 | 2013 | Isochrone Method                                           | ship performance modeling                                         | ✓               |
| Ichsan et al. (2019)        | 2019 | Energy Efficiency Operational Indicator (EEOI)             | Bottom-up approach                                                | ✓               |
| Son et al. (2020)           | 2021 | Monte Carlo, goodness of fit tests                         | Traffic probability distribution                                  | ✓               |
| Zhang et al. (2020)         | 2019 | Projection Pursuit, Info. Diffusion Theory, Risk Entropy   | Risk grading & prediction, entropy calculation                    | ✓               |
| Yoon et al. (2023)          | 2023 | Spline Interpolation, Paths segmentations, MAE             | Route Modelling                                                   | ✓               |
| Kunicka and Litwin (2019)   | 2019 | Towing tank simulator                                      | Modeling of navigational strategies                               | ✓               |
| Furuichi and Otsuka (2015)  | 2015 | Comparative Cost Analysis                                  | Structured cost estimate                                          | ✓               |
| Cai and Wen (2014)          | 2014 | Modified Isochrone Method                                  | Structured optimization & weather prediction                      | ✓               |
| Sharif et al. (2024)        | 2024 | Dijkstra, GA, Routing & MEWRA                              | Multi-objective of hard & soft constraints                        | ✓               |
| Guo et al. (2019)           | 2019 | User Equilibrium, Grey Prediction, Logit Model             | Hybrid optimization, predictive & logistic regression models      | ✓               |
| Kulkarni et al. (2024)      | 2024 | Agent-based algorithm, Discrete-Event Algorithm            | Multi-level simulation for winter navigation.                     | ✓               |
| Sasmal et al. (2021)        | 2021 | Hs-based Storm Avoidance, Monte Carlo, Prob. Distributions | Combines statistical modeling                                     | ✓               |
| Roh (2013b)                 | 2013 | Improved Isochrone, A* Algorithm, ECMWF                    | Classical optimizations use data-driven pattern recognition       | ✓               |
| Cai et al. (2021)           | 2022 | Isoline & Position Line Theory, Parallel Indexing          | Deterministic rule-based navigation model                         | ✓               |
| Hildeman et al. (2020)      | 2020 | SPDE, Matérn covariance function                           | Deformation method (from a stationary to a non-stationary domain) | ✓               |
| Panigrahi and               | 2008 | Dijkstra, WAM, Jame's Empirical,                           | Classical Optimization                                            | ✓               |

**Table 1 (continued)**

| Ref.                   | Year | Algorithms                                        | Method                                                      | DSS Suitability |
|------------------------|------|---------------------------------------------------|-------------------------------------------------------------|-----------------|
| Umesh (2008)           |      | Haltiner's Refinement                             |                                                             |                 |
| McCord et al. (1999)   | 1999 | DP, Satellite Altimetry-Based Current Estimation  | Deterministic and Optimization-based strategies             | ✓               |
| Hoorn and Knapp (2019) | 2019 | MLREF, Monte Carlo, Binary Logistic Regression    | Classical Risk Assessment Model                             | ✓               |
| Hildeman et al. (2019) | 2019 | SPDEs, FEM, MLE, GRF, Nonstationary & Anisotropic | Classical statistics using SPDEs and Gaussian Random Fields | ✓               |

generate safe, efficient, and compliant actions for both autonomous and human-in-the-loop operations. Advancing DSS capabilities is therefore central to safer navigation and to the green, efficient development of intelligent ship systems.

Given this context, there is a pressing need to advance Intelligent Maritime Transport Systems (IMTS). The effectiveness of IMTS relies significantly on situational information from mainstream navigational tools and the integration of heterogeneous data for decision support. This is particularly important for developing autonomy for Maritime Autonomous Surface Ships (MASS) (Moradi et al., 2022). In Practice, navigation decisions draw on situational awareness and expert knowledge built from radar, Automatic Identification System (AIS), and Electronic Chart Display and Information System (ECDIS), which together provide a detailed picture of the operating environment. Technically, the ship route optimization is a problem of DSS. Given a fixed or candidate route, the DSS selects the ship speed, power, and ship loading profiles that satisfy Estimated Time of Arrival (ETA) constraints while minimizing cost, fuel, and emissions. The route is partitioned into segments (waypoints), and the DSS performs combinatorial optimization across sub-routes and segment settings, accounting for interactions and trade-offs. In effect, operation optimization becomes a structured DSS workflow closely related to ship-routing tasks, where overlapping solution spaces are explored, evaluated, and ranked before presenting the operator with a small set of actionable, constraint-compliant plans.

Overall, ship routing is best treated as a decision support system (DSS) problem that fuses three pillars: (1) numerical weather forecast, (2) ship performance modeling, and (3) DSS-driven optimization into a coherent, operator-centered workflow. Therefore, the DSS processes forecast data, including wind, waves, and currents, along with models that predict vessel responses. It takes into account operational and regulatory constraints, such as ETAs, route limits, and shelter options. Using optimization DSS tools to generate, evaluate, and rank feasible route speed and power profiles is important to create effective navigation plans. This integration is essential, while inadequate routing can produce cargo delays, economic losses, and elevated navigational risk. Unlike static pathfinding, the DSS operates under uncertainty and change; it assimilates updated data, propagates forecast and model errors, and reoptimizes as conditions or commercial directives, such as ETA and the destination change. Overall, effective ship routing emerges from DSS's ability to trade off by turning heterogeneous, uncertain ship trips into tailored operational plans (Lee and Cho, 2022).

The implications of Decision Support Systems (DSS) enable navigators in evaluating routing alternatives by integrating meteorological, oceanographic, and vessel-specific data. However, many existing platforms emphasize route optimization and visualization, relying on decision logic tools such as static navigational charts and historical tracks whose provenance and quality are often uncertain, limiting their reliability for evidence-based decision-making (Liu et al., 2023). On the other hand, in maritime operations, credible forecasts of wave conditions directly support operational decision-making and voyage planning. This separation between forecasting and decision logic can reduce

the Decision Support System's (DSSs) overall reliability. Integrating credible model-based forecasts within DSS can therefore enhance voyage safety, energy efficiency, and emission reduction. In this regard, weather routing is best understood as a decision support system (DSS) that transforms forecasts and ship performance data into a small set of defensible, operator-ready options. Rather than a single optimal answer, classic and newer algorithms alike serve this role by handling stochastic inputs, dynamic constraints, and multi-objective trade-offs, then presenting navigators with transparent alternatives while retaining the human in the loop for selection (Lee and Cho, 2022). DSS, using modern approaches of data processing and optimization, predicts candidate routes, quantifies their risks and costs, and proactively filters out unfavorable outcomes. In practice, a shorter planning horizon with active route control typically reduces fuel use and emissions by avoiding adverse seas and smoothing engine loads. Although quantifying the gain is vessel and voyage-specific and beyond this study's scope, it can be computed after the fact from the chosen route's speed-power profile and the realized metocean conditions.

Recent studies have shown that Decision Support Systems (DSSs) for ship operations are implemented as a toolchain that generates and compares practical courses of action rather than a single monolithic system (Sun et al., 2022). On board the vessels, such tools address various functions like route selection, speed scheduling, sheltering choices, or weather risk screening, each driven by dedicated software modules and algorithms. Similarly, here we nominate classical time-series models (ARIMA, ARIMAX, SARIMA, VAR) as the DSS engines for short-term sea state prediction. Their forecasts parameterize a deterministic routing and scheduling module that evaluates candidate options under explicit safety and operational constraints, and returns a compact set of actionable alternatives for the operator's final choice. Thus, the perception of DSS is not a single "black box," but an integrated use of models and rules that streamline decisions over a specific operational time span. In this study, we use classical series models to extract actionable knowledge in an attempt to implement a rule-based decision-making algorithm within a decision support framework.

### 3. Analytical methods

#### 3.1. Stage 1: spectral analysis for wave height characterization

The first stage of the spectral analysis technique for wave height distribution employs spectral moment analysis, using equations 1–37 to derive key wave parameters such as significant wave height, energy density, wavelength, and peak period directly from the spectral density function. This analytical approach is fully reproducible, as each computed variable follows deterministic formulations based on angular frequency and spectral integration. By applying the same input spectrum and frequency intervals, identical outputs can be consistently regenerated, ensuring methodological transparency and repeatability in wave characterization.

#### 3.2. Stage 2: time series forecasting using EViews software

EViews software was utilized for the analysis of stage two. Its concise scientific function focused on the capabilities for time series analysis, forecasting, and modeling, with a note on reproducibility. EViews software provides a user-friendly, object-oriented interface built for statistical and econometric analysis. According to its forecast and model simulations, it supports ARIMA, ARIMAX, SARIMA, VAR, and other classical time series estimation forecasting methods. EViews software emphasizes lags, differences, seasonality, trends, and serial correlation. Also, EViews software features highlight comprehensive time-series handling with both regular and irregular frequencies (hours/minutes/seconds), and features such as automatic frequency conversion, smoothing (Holt-inters/ETS), band-pass filtering, and seasonal adjustment (Census X-13, STL). Additionally, EViews software provides simple

data handling features, including flexible import/export across formats, such as Excel, SPSS, SAS, Stata, raw text, and ODBC. This support for a large range of work files makes it suitable for univariate and multivariate forecasting. EViews supports reproducible research workflows through its command language and enables the use of EViews code from within R Markdown or Quarto documents, thereby embedding EViews commands in broader reproducible documentation environments.

### 4. Analytical dataset

#### 4.1. Dataset description

The analysis employed observational data from the National Data Buoy Center (NDBC) stations. Waves are derived from 20-min burst records; winds and met variables are reported hourly per NDBC conventions.

This study specifically examines how fluctuations in wave patterns influence ship routing decisions. It utilizes historical data from the *National Data Buoy Center (NDBC)*, which operates under NOAA's National Weather Service (NWS) as a specialized division within the US National Oceanic and Atmospheric Administration (NOAA). NDBC is NOAA's operational branch that manages and ensures the quality of buoy-based marine and atmospheric data used for weather and ocean monitoring worldwide. NDBC is responsible for designing, deploying, and maintaining the nation's network of ocean buoys and coastal marine observation systems. These buoys collect real-time meteorological and oceanographic data (winds, waves, pressure, temperature, currents) used by NOAA's forecasting, research, and warning programs. As of 2025, the NDBC manages and quality controls data from approximately 1357 ocean and coastal buoys worldwide, as depicted in Fig. 1.

Table 2 summarizes the international standardization and quality assurance frameworks used by NDBC to ensure the accuracy, reliability, and interoperability of its buoy data (Reggiannini et al., 2024). It lists key organizations and protocols, such as NIST, WMO, QARTOD, and NCEP, along with their specific roles in sensor calibration, data formatting, quality control, and model validation. Together, these standards guarantee that NDBC measurements are scientifically traceable, globally compatible, and suitable for operational forecasting and research applications.

These buoys form a global observation network that continuously records meteorological and oceanographic parameters, such as wind speed, wave height, sea temperature, air pressure, and currents. They are strategically distributed across the world's oceans, densely located near coastlines and sparser across open seas. This network provides vital real-time data that supports weather forecasting, marine safety, tsunami warning, and climate research, making it one of the most comprehensive ocean monitoring systems in operation (Tu et al., 2018). The station 41049 is used in this study, which is located south of Bermuda at approximately 31.92° N, 69.59° W. This station records key hydrometeorological variables, where data are collected at 1-h intervals and are publicly accessible via the NDBC archive. This particular station, 41049, uses the SCOOP payload, referring to NDBC's self-contained Ocean Observing Payload as a modular, sealed electronics sensor package for moored buoys. It consolidates met/ocean sensors, onboard processing, power, and comms into swappable "smart" modules, making buoys lighter, easier to service, and more reliable. SCOOP commonly rides on 3-m foam hulls and supports recording fourteen parameters: (1) WDIR (wind direction, from true North), (2) WSPD (wind speed,  $\text{m s}^{-1}$ , 8-min buoy average), (3) GST (peak gust,  $\text{m s}^{-1}$  over 5–8 s), (4) WVHT (significant wave height, m over a 20-min sample), (5) DPD (dominant wave period, seconds), (6) APD (Average wave period, seconds), (7) MWD (dominant wave direction, from true north), (8) PRES (sea-level pressure, hpa), (9) ATMP (air temperature, °C), (10) WTMP (sea-surface temperature, °C), (11) DEWP (dew point, °C), (12) VVIS (visibility, nautical miles), (13) PTDY (3-h pressure tendency, hpa), and (14) TIDE (water level, ft relative to MLLW).

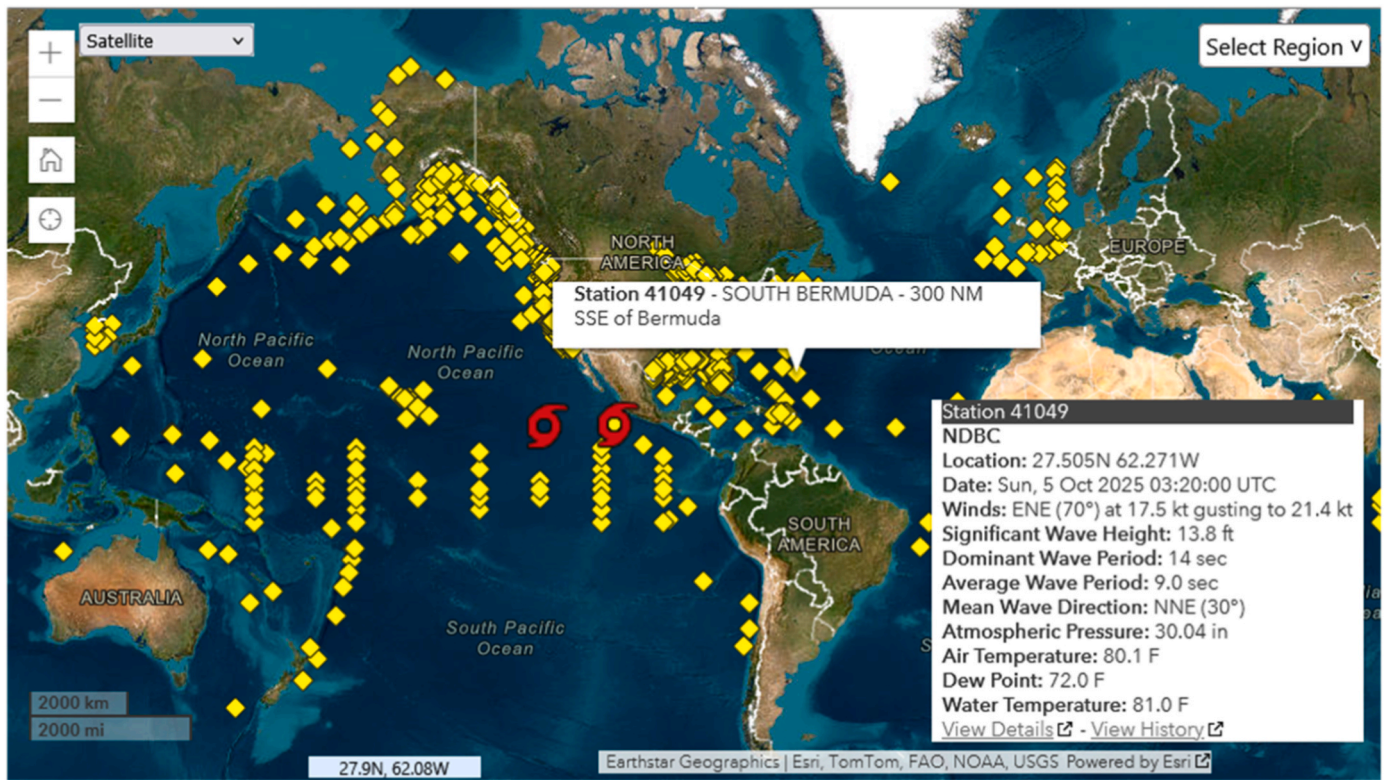


Fig. 1. Data Buoy location: Station 41049 – south Bermuda – 300 NM SSE of Bermuda.

Table 2

International Standardization and Quality Assurance Frameworks used by NDBC.

| Standardization                           | Acronyms                                                                     | Description                                                                                                           |
|-------------------------------------------|------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|
| NIST-Traceable Calibration Standard       | NIST: National Institute of Standards and Technology                         | Certified NDBC sensors to ensure measurement traceability and accuracy                                                |
| WMO Global Telecommunication System (GTS) | GTS: Global Telecommunication System                                         | Buoy data are formatted and transmitted in compliance with WMO data exchange standards for the GTS.                   |
| QARTOD (IOOS/IOC-UNESCO Standard)         | QARTOD: Quality Assurance of Real-Time Oceanographic Data                    | Automated Quality Control (QC) tests at NDBC based on QARTOD                                                          |
| NCEP/WMO Model Cross-Validation           | NCEP: National Centers for Environmental Prediction                          | Every 6 h, the measured data is compared against the numerical output of the NCEP Global Forecast System (GFS) model. |
| Independent NIST-traceable laboratory     | Independent NIST: Independent National Institute of Standards and Technology | An independent NIST test laboratory to calibrate test equipment on a regular basis.                                   |

Referring to the above discussion, the spatiotemporal data from these buoys measure environmental variables at their fixed location. After roughly two weeks, the weather and sea conditions at a certain buoy location will have changed significantly, but more importantly, the ship using its data will likely have moved into another climatic or oceanographic zone. Therefore, the buoys' record no longer represents the environment the vessel will realistically encounter. The dynamic Ship Position is a moving platform, often traversing hundreds or thousands of nautical miles. Within two weeks, a vessel can easily move from

a temperate zone into a subtropical or tropical zone, where the governing wind, current, and wave regimes differ completely. Long-term prediction would need upstream data from other buoys located along the updated route of the ship. In addition, in atmospheric science, the deterministic weather forecast's horizon has a limitation, losing skill beyond two weeks and diverging rapidly from reality [49–50]. Therefore, climate and sea state while the ship is moving between regions invalidate the assumptions underlying a single buoy's time series (stationarity) where local forcing conditions differ, and render.

#### 4.2. Data-driven first stage of analysis

Time series data set of wave spectrum density  $S_{\zeta}(\omega W)$  ( $m^2/sec$ ) and frequency  $\omega W$  ( $rad/sec$ ) were gathered from the National Data Buoy Centre (NOAA) belonging to the station 41049 South Bermuda ( $27^{\circ}30'17''$  N  $62^{\circ}16'14''$  W) – 300 NM SSE. The adopted wave spectral data is presented as a function of angular frequency ( $\omega$ ) in radians per second ( $rad/sec$ ), ranging from 0.02 to 0.14, with frequency intervals ( $\delta\omega$ ) between 0.005 and 0.02. Each entry lists the spectral density  $S_{\zeta}(\omega W)$  expressed in square meters per second ( $m^2/sec$ ), which quantifies the wave energy distribution per unit frequency. The data are structured in tabular form, where  $\omega$  defines the horizontal axis of the spectrum and  $S_{\zeta}(\omega W)$  represents the corresponding energy magnitude for each frequency component. 1980 data points were computed in this study. The first part of the analysis of wave frequency focuses on defining the surface profile to visualize wave behavior (Xu et al., 2025). The full set of used variables is presented in equations (1)–(37), section 6, calculating  $m_0, m_1, m_2, m_3, m_4, H_s, k, L_w, \zeta_w, E, \delta\omega, \omega, T_0, T_1, T_p, T_Z, T_C, NO, C, \gamma, S_B\zeta$ , and  $S_J\zeta$  provides a comprehensive description of wave characteristics.

#### 4.3. Data-driven second stage of analysis

A dataset consisting of 660 data points for the year 2023 was utilized to develop a single variable employing ARIMA, ARIMAX, and SARIMA

models. Additionally, a dataset of  $660 \times 3 = 1980$  data points (660 for one dependent variable and 660 for each of the other two independent variables) was utilized in the VAR model. This approach reinforces the appropriateness of the selected dataset for the application (Nielsen and Stredulinsky, 2012). This dataset was used to ensure adequate representation of sea wave variability. The focus variable, significant wave height (Hs), serves as a proxy for sea-state intensity relevant to the ship destination. Exogenous variables such as wind speed and Sea-Surface Temperature (SST) were incorporated in the ARIMAX and VAR models to capture environmental coupling effects.

#### 4.4. Data pre-processing and quality control

Prior to modeling, all records underwent rigorous quality control to eliminate erroneous or missing observations. Outliers were identified using an interquartile range (IQR) filter ( $\pm 1.5 \times \text{IQR}$ ) and cross-checked against the NDBC flag codes. Missing hourly data representing less than 3 % of the total were linearly interpolated; longer gaps were excluded to preserve temporal continuity.

The stationarity of the time series was assessed using the Augmented Dickey–Fuller (ADF) test. Non-stationary series were differenced as required to achieve a stable mean and variance. All variables were normalized to zero mean and unit variance prior to model calibration.

#### 4.5. Data forecasting framework

The modeling framework comprised four classical time-series approaches, selected for their analytical transparency and suitability for operational implementation:

5. ARIMA (p,d,q): captures autocorrelation and trend structure within a univariate stationary process.
6. SARIMA (p,d,q)(P,D,Q)m: extends ARIMA to incorporate seasonal periodicity m (hours = 24, 168).
7. ARIMAX (p,d,q): integrates exogenous predictors (wind speed, SST) into the ARIMA framework.
8. VAR (p): models' multivariate interdependence among wave, wind, and temperature variables.

Model identification followed the Box–Jenkins methodology. Parameters (p,d,q) and (P,D,Q)m were determined using autocorrelation (ACF) and partial autocorrelation (PACF) diagnostics combined with the corrected Akaike Information Criterion (AICc). Model adequacy was validated through residual analysis (Ljung–Box Q-test) and by ensuring residuals followed white-noise characteristics.

#### 4.6. Data forecast evaluation

Model performance was evaluated under two forecasting conditions:

- Static forecasting, where models are trained on a fixed data window and validated.

- Dynamic forecasting, where models are recursively updated as new observations become available.

Wave spectrum density  $S_{\zeta}(\omega_w)$  ( $m^2/sec$ ) and frequency  $\omega_w$  ( $rad/sec$ ) were computed based on the Time series data frame collected from the station 41049 South Bermuda ( $27^{\circ}30'17''$  N  $62^{\circ}16'14''$  W). As shown in Fig. 2, the dataset was divided into training data and testing data, comprising 1320 observations. The training data was used to build and calibrate the forecasting models, while static forecasting was performed within the same period to validate model accuracy against known values. In contrast, dynamic forecasting extended beyond the training set into the testing period, generating predictions for unforeseen time series data. This approach enables the evaluation of the model's performance both within the sample (static) and beyond the sample (dynamic), reflecting real-world forecasting scenarios.

The first part of the analysis in this study utilizes the wave spectra data set to derive a univariate time series analysis as a single-variable problem, focusing on a single sequence of observations over time (Guo et al., 2023). The second part of the analysis involves univariate and multivariate time series forecasting and incorporates multiple concurrently observed variables gathered from part one. This second part of the analysis utilizes EViews statistical software for time series analysis, forecasting, and econometric modeling. This software offers an intuitive interface that facilitates work with both univariate and multivariate data. One of its features is the ARIMA, ARIMAX, SARIMA, and VAR functions, which excel in forecasting time series data from a classical perspective. This capability enables the generation of accurate predictions based on specific historical data. The following statistical indicators were used to assess predictive accuracy:

- Root Mean Square Error (RMSE)
- Mean Absolute Error (MAE)
- Mean Absolute Percentage Error (MAPE)
- Coefficient of Determination ( $R^2$ )

Each metric provided complementary insights into model precision, bias, and reliability for short-term operational forecasting.

#### 4.7. Methodology flowchart

The modeling procedure followed the workflow summarized in Fig. 3:

- 1) Data acquisition: NDBC 41049 (South Bermuda), hourly records.
- 2) Date cleaning and quality control: filtering, interpolation, stationarity checks.
- 3) Model configuration: ARIMA, SARIMA, ARIMAX, VAR: parameters tuned via AICs.
- 4) Forecasting: static and dynamic simulations.
- 5) Evaluation: RMSE, MAE, MAPE, sMAPE, U1, U2.
- 6) Operational interpretation: integrate validated forecasts into ship-routing decision scenarios.

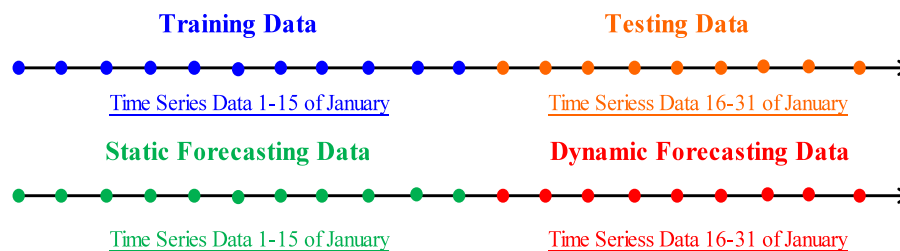


Fig. 2. Time series data segmentation for model training and forecast evaluation.

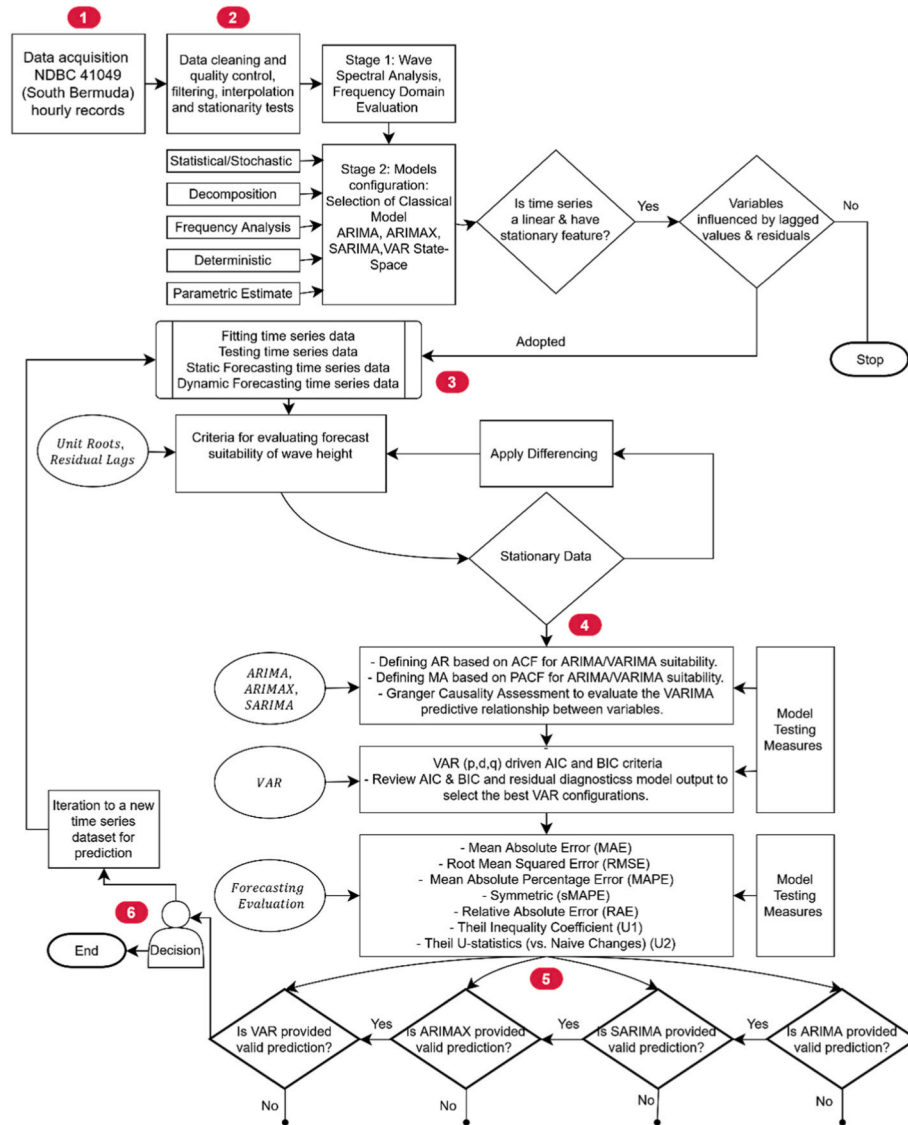


Fig. 3. Flowchart of Analytical Selection, fitting, testing, and forecasting.

This workflow provides analytical transparency and reproducibility. By separating forecasting logic from route-optimization routines, it establishes a rigorous foundation for subsequent DSS integration.

## 5. Ship optimization route

The efficiency of a ship in fulfilling its intended mission can be limited by the sea conditions. This limitation may arise from the system's ability to operate effectively and safely (Du et al., 2021). Key seakeeping parameters are commonly used to limit the impact of ship motion in waves through frameworks to improve the ship's operational capabilities under various sea conditions. The term seaworthiness encompasses all aspects of a ship's design that influence its ability to operate safely and effectively in all sea conditions while fulfilling its intended purpose. This includes considerations of endurance and factors more directly impacted by wave interactions (Weintrit and Neumann, 2011). In this case, attention is focused on head seas where the term seakeeping specifically refers to features such as motions, speed, power in waves, wetness, and slamming. In calm water, a ship traveling at constant power will maintain a constant speed. However, when encountering waves, the speed decreases on average due to added resistance and altered operating conditions for the propeller. In these situations, the speed is not

constant; instead, it becomes variable. The term “surge” or “surge velocity” describes the fluctuations in the vessel's speed around this new average. Thus, an irregular sea of wave heights and resistance varies, causing irregular fluctuations in the ships' speed. These sea irregularities of waves are inconsistent and directly influence the vessel's performance to become highly nonlinear and complex to predict (Lee et al., 2022). The relative significance of these seakeeping features varies depending on the operational needs of the ship. However, the optimum principle of seakeeping is relevant to most vessels to provide valuable guidance for achieving optimal operational performance in wave conditions. While good seakeeping is undeniably desirable, the challenge lies in balancing it with other operational and design features. The extent to which these features should be compromised to enhance seakeeping depends on the specific operational requirements. Thus, it is crucial to have reliable methods to evaluate the expected seakeeping performance and its impact on the ship's overall effectiveness. In a competitive industry, a ship that offers greater economic service is more likely to derive a cost benefit rather than poor performance. Similarly, a vessel that experiences less power augmentation in waves can maintain tighter schedules or reduce fuel consumption. Especially in extreme conditions, a ship's seakeeping qualities determine its ability to complete a voyage economically.

Given the random and unpredictable nature of the sea, statistical analysis plays a significant role in assessing seakeeping performance. Fuel consumption in ships is significantly influenced by hydrodynamic and aerodynamic resistance generated by waves, wind, and ocean currents. These external forces increase the effective power demand on the propulsion system, thereby raising fuel usage. By optimizing vessel speed and heading in relation to prevailing environmental conditions, resistance can be reduced, resulting in lower engine loads and improved fuel efficiency. Therefore, fuel efficiency and carbon emission reduction have become critical priorities in modern naval operations.

We acknowledge the observation regarding fuel consumption and clarify that our study does not attempt to compute fuel use directly because fuel expenditure is a multifactorial outcome that depends on ship type, hull form, displacement, loading condition, propulsion efficiency, fuel type, and voyage duration. Quantifying fuel consumption requires vessel-specific propulsion and engine performance models, which lie outside the scope of the present work. Instead, our contribution focuses on forecasting the sea-state energy field, the primary external driver of added resistance and speed loss. By predicting significant wave height, dominant period, and spectral energy through reproducible time-series models, we generate the environmental inputs required by operational routing tools or ship-specific energy models to estimate fuel demand under varying conditions. Thus, our forecasting framework acts as a predictive decision-support layer that enables route planners and onboard systems to anticipate adverse wave conditions, adjust speed or heading proactively, and maintain propulsive efficiency. In this way, the method indirectly supports fuel optimization without requiring any vessel-specific or engine-specific parameters. Future work will integrate these forecasting outputs with ship-resistance and propulsion models (e.g., Holtrop-Mennen or CFD-based EEDI methods) to quantify absolute fuel-saving potential for selected vessel classes.

Achieving an optimal ship route, defined as the path of minimum fuel consumption, depends not only on vessel-specific characteristics but also on near-real-time environmental factors such as weather, ocean waves, and currents. The environmental footprint of maritime operations can be evaluated across multiple spatial and temporal scales. Within this context, notable examples include the U.S Navy's smart voyage planning (SVP) decision aid, which integrates the meteorological and oceanographic (METOC) forecast system to support routing decisions (Moradi et al., 2022). Similarly, advanced DSS frameworks employing Artificial Neural Network (ANN) models provide onboard, real-time predictions of fuel consumption using operational inputs, such as speed, RPM, draft, trim, cargo load, and sea state. These systems outperform traditional regression models, enabling more informed, cost-saving operational decisions (Nielsen and Dietz, 2020). Moreover, DSS design uses adaptive incremental learning to improve fuel consumption predictions over time, supporting better planning and fuel optimization (Lee and Cho, 2022). Such adaptive routing strategies not only enhance operational safety and seakeeping performance but also contribute to the reduction of greenhouse gas emissions and compliance with international maritime sustainability targets.

Beyond improving a ship's physical response to waves, its overall effectiveness can also be enhanced through the strategic placement of critical operations and the installation of control and decision-making devices like *decision support systems (DSS)* for route optimization. At the same time, the *International Maritime Organization (IMO)* has adopted key measures to reduce *Greenhouse Gas (GHG)* emissions from ships, targeting net-zero emissions by around 2050. This goal is outlined in the 2023 *IMO* strategy under resolution *MEPC. 377(80)*, which evolved over several sessions, including meetings from July 2023 to April 2025. The agreement led to amendments in *MARPOL Annex VI*, introducing a *GHG* fuel Standard, effective in March 2027. The aim is to reduce annual *GHG* emissions from international shipping by at least 20 % by 2030, compared to 2008 levels, 30 %, extending to 70 % by 2040, with *NZE* expected by 2050 (Panigrahi et al., 2008). In this context, the integration of decision support systems (*DSS*) becomes even more critical, not

only for improving navigational safety and operational reliability but also as a mechanism to support *GHG* reduction goals. A ship's freedom of action is often constrained by seakeeping limitations. Traditionally, this leads to speed alteration to avoid resonance with wave frequencies or course alterations when wave directionality allows.

However, the technology of *decision support systems (DSS)* lies in its ability to dynamically analyze predicted wave conditions and adjust optimal routes within a polar plot framework. By integrating long-term wave forecasts and heading adjustments, the objective of *DSS* is to offer actionable insights for navigators to optimize voyage planning. This enables vessels to maintain energy-efficient speeds and safe direction, reducing fuel consumption and gas emissions while mitigating risks related to crew fatigue and cargo safety. Ultimately, *DSS* aligns with both the ship's operational efficiency and reliability of seakeeping and the environmental conditions of international maritime policy. The larger the sea area, the broader the range of speeds and headings in which the vessel can operate effectively. In all case scenarios, *DSS* ensures that vessels can operate effectively in an optimal course in varying sea states.

To proceed with the time series analysis, we first summarize the dataset used for fitting and testing. Fig. 3 displays the time series data of significant wave height utilized in this study via four forecasting models: *ARIMA*, *SARIMA*, *ARIMAX*, and *VAR*. The sample of time series data from *January 2023 (Hs-23-January)* is used for training and fitting the model. We expect that the predictions generated from the training sample *Hs-23-January* of the first 15 days forecast the subsequent dynamics of the next 15 days, as influenced by similar spatiotemporal factors. The challenge of using *ARIMA*, *SARIMA*, *ARIMAX*, and *VAR*-based models is whether these models are able to provide a minimum prediction for two weeks' time. Additionally, we will explore the extent to which these models can capture and predict the dynamics of time series data. The core challenge examined in this study is whether these classical models can effectively capture trends, short-term fluctuations, or both within the wave height of time series data. Additionally, the study investigates whether these predictive capabilities are sufficient to support *Decision Support Systems (DSS)* for safe and efficient ship routing.

## 6. Dynamics of sea wave spectrum

Understanding the spectral density function is essential for analyzing the energy distribution across various frequencies, which is critical for interpreting wave dynamics and assessing their impact on structures and shorelines. In general, the inherent complexity driven by the spatial and temporal variability of waves makes predicting and harnessing wave energy a challenging task. The wave spectrum reveals that wave system energy tends to concentrate around a specific frequency, with minimal energy at both lower and higher frequencies. The full set of variables is presented and calculated via  $m0, m1, m2, m3, m4, H_s, k, L_w, \zeta_a, E, \delta\omega, \omega, T0, T1, T_p, T_z, T_c, N0, C, \gamma, S_{B\zeta},$  and  $S_{\zeta}$  provides a comprehensive description of wave characteristics. However, only a subset was applied selectively for modeling purposes. Specifically, three variables were incorporated into the *VAR* model, whereas a single variable was employed for *ARIMA*, *SARIMA*, and *ARIMAX*. The rationale for this selection, as well as its capture of the essential dynamic implications for operational ship routing, is discussed in the following sections. This selective use of variables provides the foundation for the subsequent analysis, which shifts toward examining multivariate influential features and their role in predicting ship response through time series models.

These wave spectrum variables were processed using an *ARIMA* model to support a decision support system (*DSS*) on board a vessel for predicting ship response direction effectively. The dataset includes a range of each wave frequency ( $\delta\omega$ ), wave number ( $k$ ), and wavelength ( $L_w$ ), representing a temporal phenomenon characterized by wave period and wavelength. Higher frequencies indicate shorter wave periods and wavelengths, resulting in choppy waves that dissipate energy more quickly. Conversely, lower frequencies correspond to longer wave

periods and wavelengths, producing larger swells capable of carrying energy over greater distances. The density spectrum reflects the energy distribution and peaks at specific frequencies, but its role varies depending on the source of wave energy, such as air, wind, and water currents, all influenced by environmental factors. The phenomena of energy dissipation act oppositely to wave generation, and the relationship between wave frequency and density spectra is complex due to the dynamic nature of oceanic processes. At low frequencies (longer waves), energy declines rapidly as smaller, shorter-period waves diminish quickly. The spectral density function quantifies how wave energy is distributed across different frequencies. It provides valuable insights into the dominant wave systems within a specific frequency band, influenced by factors such as wind conditions and fetch length. Wave frequency and wave period are inversely related, with higher frequencies corresponding to shorter periods and wavelengths. The time between successive wave crests, or wavelength, decreases as frequency increases.

This phenomenon of wave motion displays a long-crested sea behaviour, characterized by the movement of waves from their crests and the direction of propagation (Harris et al., 2018). This 2D behaviour forms part of the broader 3D wave spectrum, often referred to as the short-crested 2D spectrum. The wave spectrum is an expression of wave behaviour at specific locations, enabling the formulation of predictive models tailored to particular spatiotemporal conditions at sea. The fundamental property of ocean waves is their speed ( $C$ ), which is proportional to their wavelength and is also represented by the wave period ( $T$ ) per second. Longer waves travel at higher speeds, while shorter waves move at lower speeds. Despite their apparent, with minimal net mass transport. Specifically, water moves forward at wave crests and backward at troughs. The significant wave height ( $H_s$ ) is a key parameter, defined as the average height of the highest one-third of waves ( $H_{1/3}$ ). For non-breaking waves,  $H_s$  is approximately equal to  $H_{m0}$  where  $H_{m0}$  denotes the spectral significant wave height derived from wave energy, as depicted in equation (1). Ocean waves, being much denser than air water is 850 times denser), carry substantial energy. However, this energy diminishes as waves enter shallow water, where the depth is less than half the wavelength. At this stage, wave energy dissipates due to friction with the seabed and ultimately dissipates as waves break onshore. Wave energy is quantified as energy flux or energy flow, expressed as the average power per meter of wave crest length (W/m). While wave energy is dependent on wave amplitude, it is independent of wave frequency. The wave frequency spectrum provides information about the energy, attitude, and its distribution across different frequencies. This involves a two-dimensional analysis of data represented through variables denoted in  $m0$ ,  $H_s$ ,  $k$ ,  $\zeta_a$ ,  $\delta\omega$ ,  $\omega$ , and  $S_\zeta$ . The variables of the wave spectrum can be mathematically represented below by the equations from 1 to 6. The concept of wave spectrum and energy spectrum shares the same ordinates, where the wave spectrum, often referred to as the spectral density of wave energy, is defined by its spectral ordinates for individual frequency bandwidths and can be defined by dividing  $\frac{1}{2} \zeta_n^2$  as denoted in equations (2) and (3).

Time series data of wave frequency consists of numerical parameters that can generally be categorised into descriptive modelling to analyze and understand the dataset's behaviour. In this study, the primary focus is on predictive modelling for time series forecasting. However, descriptive modelling plays a crucial role in identifying the behaviour of data variable components using the equations denoted from 1 to 6. The outcomes of this initial analysis will serve as input data, containing designated variables for the subsequent stage of multivariate analysis.

$$V_w = \frac{L_w}{T_w} = \frac{g}{\omega_w^2} = \left( \frac{g l_w}{2\pi} \right)^{1/2} \quad (1)$$

$$L_w = \frac{2\pi V_w^2}{T_w} = \frac{2\pi g}{\omega_w^2} = \frac{g T_w^2}{2\pi} \quad (2)$$

$$T_w = \left( \frac{2\pi L_w}{g} \right)^{1/2} \quad (3)$$

$$P = g \rho z \pm \zeta_a^2 g \rho e^{-kz} \cos(kx - \omega_w t) \quad (4)$$

$$L_w = \frac{2\pi}{L_w} = \frac{\omega_w^2}{g} = \frac{g}{V_w^2} = \frac{4\pi^2}{g T_w^2} \quad (5)$$

$$\alpha_M = k \zeta_a = \frac{2\pi \zeta_a}{L_w} = \frac{\pi h_w}{L_w} \quad (6)$$

$$E = 1/2 \rho \zeta_a^2 \quad (7)$$

$$H_s = 4 \times \sqrt{m0} \quad (8)$$

$$S_\zeta(\omega)_{wn} = \left( \frac{1}{2} (\zeta_n^2) \right) / \delta\omega_w \quad (9)$$

$$m0 = \int_0^\infty S_\zeta(\omega_w) \delta\omega_w \quad (10)$$

$$\delta\omega_n = \delta\omega_{n+1} - \delta\omega_n \quad (11)$$

$$k = \frac{\omega^2}{g} \left( \frac{\text{rad}}{\text{meter}} \right) \quad (12)$$

$$l_w = 2 \frac{\pi}{k} (\text{meter}) \quad (13)$$

$$\zeta_a = 2 \times \sqrt{\delta\omega \times S_\zeta(\omega_w)} \quad (14)$$

$$S_{\zeta(\omega_w, \theta)} = \zeta_a \cos(kx - \omega_t) (\text{meter}) \quad (15)$$

$$E_T = 1 / 2 \rho g (\zeta_1^2 + \zeta_2^2 \dots + \zeta_n^2) / \delta\omega \quad (16)$$

$$m0 = \delta\omega \times S_\zeta(\omega_w) \quad (17)$$

$$m1 = \delta\omega \times S_\zeta(\omega^1 w) \quad (18)$$

$$m2 = \delta\omega \times S_\zeta(\omega^2 w) \quad (19)$$

$$m3 = \delta\omega \times S_\zeta(\omega^3 w) \quad (20)$$

$$m4 = \delta\omega \times S_\zeta(\omega^4 w) \quad (21)$$

$$T0 = \frac{2\pi m0}{m1} \quad (22)$$

$$T1 = 2\pi \sqrt{\frac{m0}{m1}} \quad (23)$$

$$Tz = 2\pi \sqrt{\frac{m0}{m2}} \quad (24)$$

$$Tc = 2\pi \sqrt{\frac{m2}{m4}} \quad (25)$$

$$Tp = T0 = \frac{2\pi}{\omega p} \quad (26)$$

$$N0 = 1 / 2\pi \sqrt{\frac{m2}{m0}} \quad (27)$$

$$S_{B\zeta1}(\omega_w) = A / \omega_w^5 \times \exp(-B / \omega_w^4) m^2 / (\text{rad} / \text{sec}) \quad (28)$$

$$S_{B\zeta2}(\omega_w) = A / \omega_w^5 \times \exp(-B / \omega_w^4) m^2 / (\text{rad} / \text{sec}) \quad (29)$$

$$B1 = \frac{33.56}{H_{1/3}^2} \quad (30)$$

$$B2 = \frac{691}{T_1^4} \quad (31)$$

$$S_{J\zeta 1}(\omega_w) = 0.658.C.S_{B\zeta 1}(\omega_w) m^2 / (rad / sec) \quad (32)$$

$$S_{J\zeta 2}(\omega_w) = 0.658.C.S_{B\zeta 1}(\omega_w) m^2 / (rad / sec) \quad (33)$$

$$J = \exp \left[ \frac{-1}{2\gamma^2} \left( \frac{\omega T_0}{2\pi} - 1 \right)^2 \right] \quad (34)$$

$$\gamma = 0.07 \text{ for } < \frac{2\pi}{T_0} \quad (35)$$

$$\gamma = 0.09 \text{ for } > \frac{2\pi}{T_0} \quad (36)$$

$$C = 3.3^J \quad (37)$$

where;

Wave Velocity ( $V_w$ ),  
Wavelength ( $L_w$ ),  
Wave Period ( $T_w$ ),  
Pressure distribution under a wave ( $P$ )  
Wave number ( $k$ ),  
Wave Height ( $H_s$ )  
Spectral moments: ( $m_0, m_1, m_2, m_3, m_4$ )  
Wave periods overall wave climate: ( $T_0 = T_p$ )  
Mean wave period of highest wave amplitude: ( $T_1$ )  
mean zero-crossing frequency: ( $N_0$ )  
Bretschneider spectrum: ( $SB\zeta 1(\omega_w)$ ):  
Spectrum coefficients:  $B1, B2$   
JONSWAP Spectrums:  $SJ\zeta 1, SJ\zeta 2$   
Maximum Wave slope (first approximation) ( $\alpha_M$ ),  
Wave Energy per unit area ( $E$ )  
Total spectrum energy ( $E_T$ )  
Spectral Density: ( $S_\zeta(\omega_w)$ )  
Wave Spectrum ordinates for each frequency bandwidth:  $S_\zeta(\omega_{wn})$   
Wave frequency ( $\omega_w$ )  
Bandwidth of wave frequency: ( $\delta\omega$ )  
Mean number of wave zero up-crossings per second: ( $T_z$ )

Average wave period between two neighboring wave crests: ( $T_c$ )  
interval between consecutive wave peaks/crests: ( $T_z$ )  
Peak enhancement function: ( $J$ )  
Peak enhancement factor: ( $\gamma$ )  
Amplitude: ( $\zeta_a$ )

## 7. Analysis part 1: computations of wave spectrum parameters

Fig. 4, added to clarify the sequence of energy distribution via wave spectrum across different frequencies, sheds light on prominent peak waves to indicate wave energy concentrations at particular frequency ranges. Fig. 5 shows the wave spectrum, also defined as the spectral density of wave energy, where the spectral ordinates for each bandwidth frequency are calculated using  $\frac{1}{2} \zeta_n^2$  at each frequency, as clarified in equations (2) and (3).

Several spectrum moments can be computed, particularly, the zeroth moment ( $m_0$ ) adds value to study wave energy spectrum and its areas to compute the wave significant heights ( $H_s$ ), which is the variable we are concerned with using for the year 2023 for fitting and testing. The continuous line plots in Figs. 6 and 7 display the maximum  $Om$  and  $H_{s-23}$  in relation to energy spectrums per time slot of wave height, as illustrated in Fig. 5.

Maximum values of the results were chosen to interpret the influence of the maximum  $Om$  values to generate maximum values of significant heights  $H_s$ . This detailed frequency distribution captures wave behaviour over a broad frequency spectrum. The wave frequencies were introduced in radians and gradually increased in steps of approximately 0.005–0.02 rad/s across 47 data points from incrementally lower to larger range. The results in Table 3 correspond to the maximum  $Om$  at all frequency points, which is crucial to calculating the wave's significant heights  $H_s$ .

It is essential to consider that Table 3 summarizes the zeroth moment  $Om$  as a key parameter, introducing the area under the wave spectrum, indicating the total energy distribution over January 2023,  $Om_{23}$ . The results reveal variations in maximum wave areas across different frequencies and years, showing variability and not following a strictly linear trend, which fluctuates dependently on a unique energy distribution of each frequency. The area 0.1100  $m^2$  demonstrates the non-linear nature of wave energy distribution and its inherent predictive complexity.

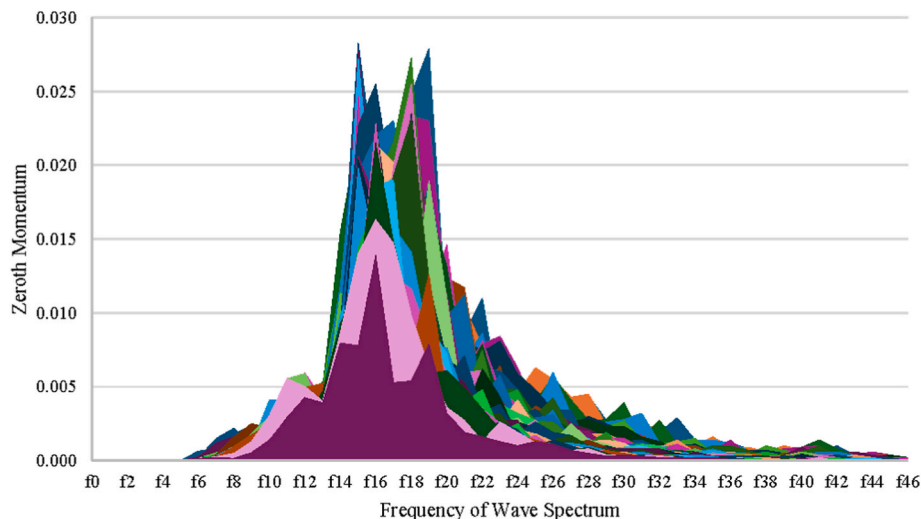


Fig. 4. Distribution of wave energy spectrum.

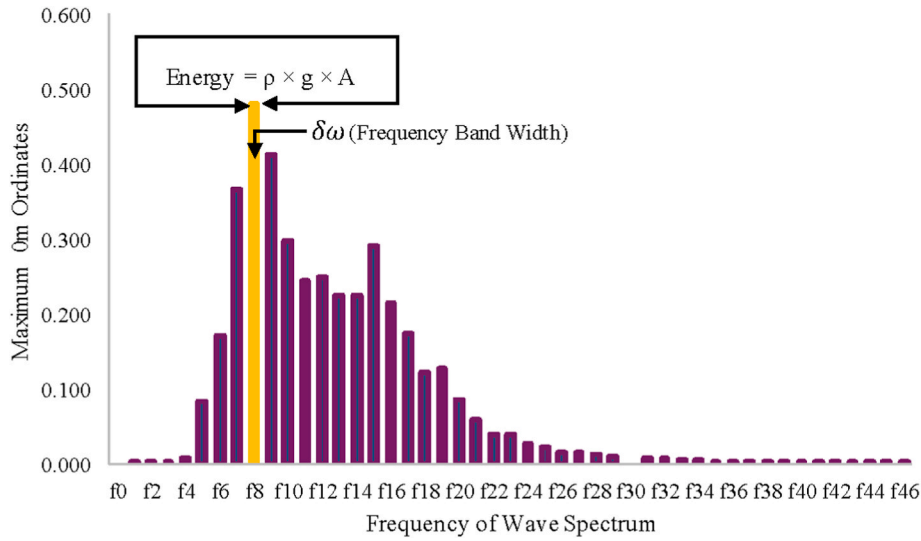


Fig. 5. Spectral ordinates of wave spectrum.

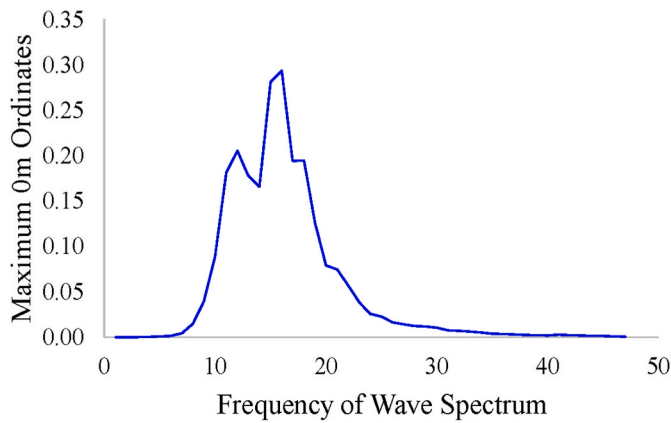


Fig. 6. Om fitting sample – Jan. 2023.

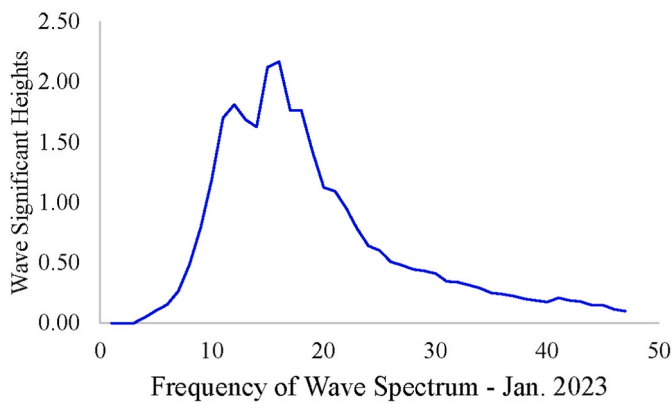
Fig. 7.  $H_{s,23}$  testing sample – Jan. 2023.

Table 3

Zeroth moment.

| Year | Wave Frequency (rad/s)<br>( $\omega_{Max}$ ) | Zeroth Moment ( $m^2$ )<br>$Om = \delta\omega \times S_{\zeta}^2$ | Significant Heights (m)<br>$H_s = 4 \times \sqrt{Om}$ |
|------|----------------------------------------------|-------------------------------------------------------------------|-------------------------------------------------------|
| 2023 | $\omega_{Max}$ 0.1100                        | $Om_{23}$ 0.2938                                                  | $H_{s,23}$ 2.1681                                     |

## 8. Analysis part B: univariate prediction of waves' significant heights

The fundamentals of forecasting are a prediction, estimate, or projection of future activity, occurrence, or event. We are focusing on quantitative forecasting methods introduced through two main pipelines: (1) *Time-series models*, which look at past patterns of data to predict future events based on the significant patterns underlying those data, while the second standard models defined as (2) *Associative Models*: Sometimes defined as causal models that the illustrated forecasted variable is related to other variables in a particular context, and its projection based on this association. The most straightforward approach within this category is a linear regression model, expressed in the form  $y_i = a \times t + b$ , where  $y_i$  is the value of the target variable at a time, and the coefficients' values  $a$  and  $b$  can be estimated to forecast the future values  $y$ . Among these techniques, the most widely recognized ones are ARIMA, ARIMAX, SARIMA, and VAR models (Hildeman et al., 2022).

### 8.1. ARIMA: Autoregressive Integrated Moving Average model

To check for the model stationarity, it is important to note that an ARMA model ( $p, q$ ) is applicable when the time series data exhibits a constant mean and variance, requiring only autoregressive (AR) and moving average (MA) components. If the time series data is not stationary, an AR(D)MA model ( $p, d, q$ ) is required, where  $d$  represents the order of integration, indicating the number of differencing steps needed, e.g., 1, 2, to achieve stationarity. In summary, when the time series data is stationary, we provide model estimates using the ARMA approach. When the time series data is nonstationary, we estimate it via the ARIMA model.

The Akaike Information Criteria (AIC) test identified the top 20 best-fitting models, which were evaluated using the *Correlogram Q-Statistics* (Ljung-Box) of Autocorrelation (AC) and Partial Autocorrelation (PAC). Among them, ARIMA (1, 1, 2) model offers the best performance in terms of simplicity and overall predictive power, yielding white noise residual diagnostics and forecast alignment with actual data. The best candidate model (ARIMA) (1, 1, 2), including 36 lags. The analysis of 36 lags revealed no clustering of high AC or PAC values, as all lag values fluctuated randomly around zero. All Q-statistic  $p$ -values  $> 0.05$  suggest the model's stability, indicating that ARIMA model (1, 1, 2) on  $H_{s,23}$  allows for valid forecasting. Proceeding to the next step of ARIMA analysis, *Series Statistics*, the *Unit Root* test was conducted using the

Augmented Dickey-Fuller (ADF) test at level 0 of the ARMA model. It is found that the  $p$ -value of the unit root test level 0 = 0.1630, and  $p$ -value > 0.05, where differencing is needed. The 1st difference of the unit root test is implemented, the  $p$ -value found is less than 0.05, and ADF  $t$ -statistic = -17.15459 is much lower than -3.4539 (1 % level),  $HS_{23}$  is now stationary.

Static forecasting of the time series data for the training sample was implemented, utilizing the estimated equation  $d(hs_{23\_jfh\_training}) \sim ar(1) ma(2)$ . The training forecasting result is compared with the actual time series data as depicted in Fig. 8. The forecast line (in red) generated by the candidate ARIMA model (1, 1, 2) aligns closely with the frequency of actual wave height values observed. This static forecast accurately mirrors the trend in the observed data line (in blue). Also, the forecast remains within a consistent range throughout most of the time series data, adjusting continuously to follow the observed data. 95 % of the data points fall within  $\pm 2$  standard errors (S.E.), indicating a 5 % confidence interval for forecasting. This result demonstrates that the model captures both the short-term fluctuations and the overall trend in wave height. Therefore, it indicates that the model is well-calibrated and suitable for predicting sea wave height conditions.

Building on these findings, we embarked on the second phase of dynamic forecasting for predicting the wave height beyond the set of time series data, utilizing the testing time series data. The univariate dynamic prediction indicated by the forecasting dynamic result line (in blue) in Fig. 9 suggests that wave heights during the 2nd half of January 2023 ranged from 0.70 m to 1.0 m. This prediction reflects the range of the significant wave heights, shown in the orange line.

In the same Fig. 9, compiling both the training and testing time series data provides a comprehensive view of how the ARIMA model captures the overall trend of significant wave height. In Particular, the transition from the training phase (black) to the dynamic forecast (blue), when compared with the testing time series data (orange). Although the dynamic forecast (blue) smooths out short-term variations, it effectively reflects the ability to follow the wave height movement in the testing sample of time series data. This forecasting result suggests that the ARIMA model is relatively capable of predicting the overall trend of wave height variations. While it may not typically mirror the natural fluctuations of wave height, the model effectively represents the overall pattern and directional movement of the time series data. The dynamic forecast captures the overall trend of wave height, which is particularly valuable in applications such as ship routing, where anticipating maximum wave heights is essential for operational decision making. For greater accuracy, especially in scenarios influenced by external factors, ARIMA can be further enhanced by seasonality via SARIMA modelling, or by integrating exogenous variables via ARIMAX modelling, or extended to multivariate analysis using models such as VAR, which we have discussed in the subsequent sections.

#### 8.1.1. ARIMA forecasting evaluation

Table 4 displays the acceptable ranges and metrics utilized for forecasting evaluation (Sun et al., 2022).

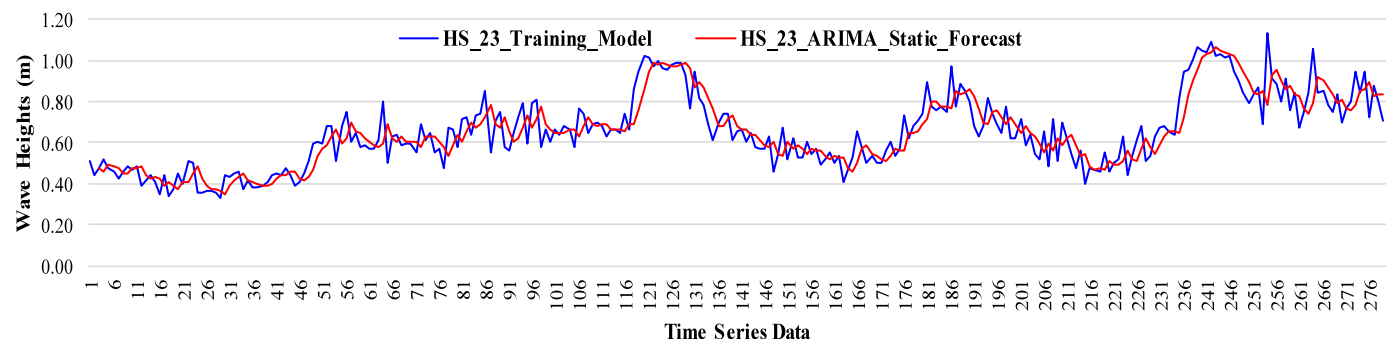


Fig. 8. ARIMA forecasting of training sample –  $HS_{23}$  January.

ARIMA forecasting result was evaluated using several error metrics (Please refer to Table 5), which represents an RMSE = 0.2798, indicating moderate forecast error and acceptable predictive accuracy. Theil U1 statistic (0.1833) close to zero indicates the model performs better than a naïve benchmark, while Theil U2 (8.7743) > 1.0 suggests poor performance in predicting changes. High MAPE (46.9037 %) and sMAPE (34.0510 %) values highlight the model's difficulty in capturing short-term fluctuations in wave height. However, the MAE (0.2382) demonstrates a stable average deviation. Overall, the error metrics indicate that ARIMA model exhibits reasonable trend forecasting performance, but it is limited in short-term accuracy. This is supported by strong Theil U1, RMSE, and MAE values, all of which are essential for decision-making.

#### 8.2. ARIMAX: ARIMA adding exogenous variables

Pressure parameters were integrated into the ARIMAX model as exogenous variables to improve predictive accuracy, under the assumption that they significantly influence wave heights. Exogenous variables are external variables not part of the time series data of the dependent variables, but helping to predict them, as shown in equation (38).

$$y_t = c + \sum_{i=1}^p \phi_i y_{t-i} + \sum_{k=1}^m \beta_k x_{t-k} + \epsilon_t \quad (38)$$

where:

- $y_t$ : the dependent variable of time series being forecasted
- $c$ : A constant term
- $\phi_i$ : Autoregressive coefficients of AR
- $\beta_k$ : the coefficients of exogenous variables
- $\theta_j$ : Moving average coefficients of MA
- $\epsilon_t$ : The error term at time  $t$
- $x_{t-k}$ : The exogenous variables

Cointegration testing was implemented on both variables,  $HS_{23}$  and  $DPD_{23}$ , to determine whether a long-run equilibrium relationship exists, as depicted in Table 6. This test regresses the original non-differenced wave height variable  $HS_{23}$  on the independent variable  $DPD_{23}$ . The result in Table 6 shows  $DPD_{23}$  coefficient value = 0.0322 and  $p$ -value = 0.0013, indicating a strong statistical relationship. This confirms a valid long-run regression, allowing for the next analytical step of ADF.

The Unit Root test via the Augmented Dickey-Fuller (ADF) test confirms that  $DPD_{23}$  is integrated at the first differenced form of order 1 ( $I(1)$ ).  $DPD_{23}$  becomes stationary after first differencing, with a highly significant test statistic (-17.15) much lower than the critical values of 1 %, 5 % and 10 % levels. In addition,  $p$ -value < 0.05, and  $R$ -squared value = 0.7402, provide a good model fit for the differenced regression. Durbin Watson stat value = 2.012, close to the value 2.0, indicating no

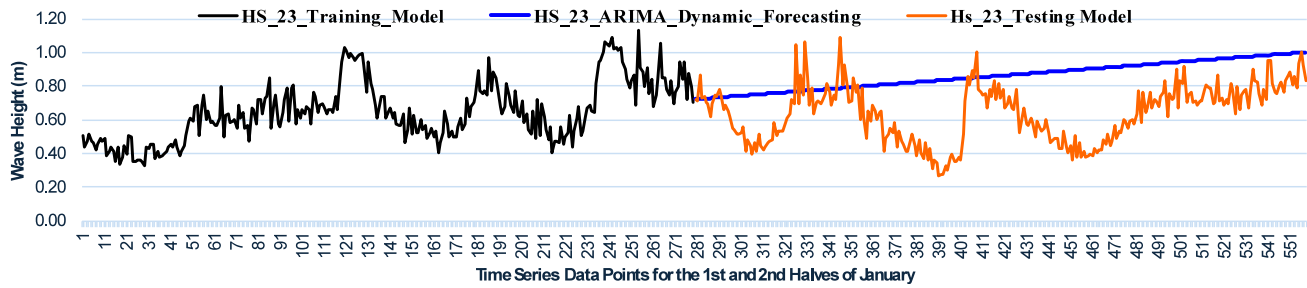


Fig. 9. ARIMA static, dynamic forecasting, testing sample – January 2023.

Table 4

Acceptable ranges for forecast evaluation metrics.

| Metric | MAE                               | RMSE                              | RAE                               | MAPE                           | sMAPE                     | Theil's U1                        | Theil's U2                             |
|--------|-----------------------------------|-----------------------------------|-----------------------------------|--------------------------------|---------------------------|-----------------------------------|----------------------------------------|
| Name   | Mean Absolute Error               | Root Mean Squared Error           | Relative Absolute Error           | Mean Absolute Percentage Error | Symmetric MAPE            | Theil Inequality Coefficient      | Theil U-statistic (vs. Naïve changes)  |
| Range  | $\geq 0$ , desired (Lower better) | $\geq 0$ , desired (Lower better) | $= 0$ , perfect ( $< 1$ , better) | $0-100+\%$ (Lower better)      | $0-100+\%$ (Lower better) | $= 0$ , perfect ( $< 1$ , better) | $< 1$ , desired ( $> 1$ , not desired) |

Table 5

ARIMA forecast evaluation for HS\_23.

| Forecast Evaluation Statistics |        |        |          |           |          |          |        |
|--------------------------------|--------|--------|----------|-----------|----------|----------|--------|
| ARIMA Forecast                 | RMSE   | MAE    | MAPE (%) | SMAPE (%) | Theil U1 | Theil U2 | RAE    |
| HS_23.Forecasting              | 0.2798 | 0.2382 | 46.9037  | 34.0510   | 0.1833   | 8.7743   | 1.6807 |

Table 6

Cointegration test.

| Variable           | Coefficient | Std. Error            | t-Statistic | Prob.     |
|--------------------|-------------|-----------------------|-------------|-----------|
| C                  | 0.382378    | 0.083497              | 4.579542    | 0.0000    |
| DPD_23             | 0.032155    | 0.009875              | 3.256384    | 0.0013    |
| R-squared          | 0.036870    | Mean dependent var    |             | 0.652152  |
| Adjusted R-squared | 0.033393    | S.D. dependent var    |             | 0.176935  |
| S.E. of regression | 0.173956    | Akaike info criterion |             | -0.652885 |
| Sum squared resid  | 8.382216    | Schwarz criterion     |             | -0.626855 |
| Log likelihood     | 93.07752    | Hannan-Quinn criter.  |             | -0.642444 |
| F-statistic        | 10.60404    | Durbin-Watson stat    |             | 0.354486  |
| Prob(F-statistic)  | 0.001269    |                       |             |           |

autocorrelation in test residuals. This justifies the inclusion of the variable  $DPD_{23}$  as a valid exogenous regressor in the  $ARIMAX$  model. Based on the  $ARIMAX$  candidate model  $(1,1,2)$ , implementing the estimated equation  $d(hs_{23}) dpd_{23} c ar(1) ma(2)$  in Table 6 demonstrates statistically significant relationships.  $AR(1)$  coefficient is negative =  $-0.5721$ , with a  $p$ -value =  $0.0000$ , indicating significant autocorrelation at lag 1, while  $MA(2)$  is a negative coefficient =  $-0.2570$ , with a  $p$ -value =  $0.0003$ , reflecting significant moving average at lag 2. Model diagnostics,

including  $Durbin-Watson\ stat = 2.0200$ , suggest the model is well specified.

The Correlogram of residuals for  $ARIMAX$  model  $d(hs_{23}) dpd_{23} c ar(1) ma(2)$  shows no significant autocorrelation up to 36 lags, with all  $Q$ -statistic  $p$ -values above  $0.31$ . This suggests that the residuals are white noise, confirming that the model sufficiently captures the data dynamics and is appropriate for static and dynamic forecasting.

In the following Fig. 10, the static forecasting of  $HS_{23}$  (black line) closely follows the training time series data (red line), accurately capturing both the trend and fluctuations in wave height over the first half of January. This alignment indicates that the  $ARIMAX$  model provides reliable short-term predictions within the training period.

In Fig. 11, the dynamic forecasting graph (blue line) varies between  $0.4$  m and  $0.95$  m, showing a general upward trend. It partially captures the variability of testing data (orange line). Notably, testing time series data remains within a wider range of about  $0.2$  m– $1.0$  m. This suggests that while  $ARIMAX$  model tracks overall trends and indicates a reasonable range of wave height, it struggles to follow the fluctuation of sea waves.

#### 8.2.1. ARIMAX forecast evaluation

The  $ARIMAX$  model was assessed using error metrics in Table 7. These metrics highlight its improved performance compared to  $ARIMA$ . The result reports an  $RMSE$  ( $0.1403$ ), reflecting a relatively low forecast error and improved predictive accuracy.  $Theil\ U1$  statistic ( $0.1076$ ) is

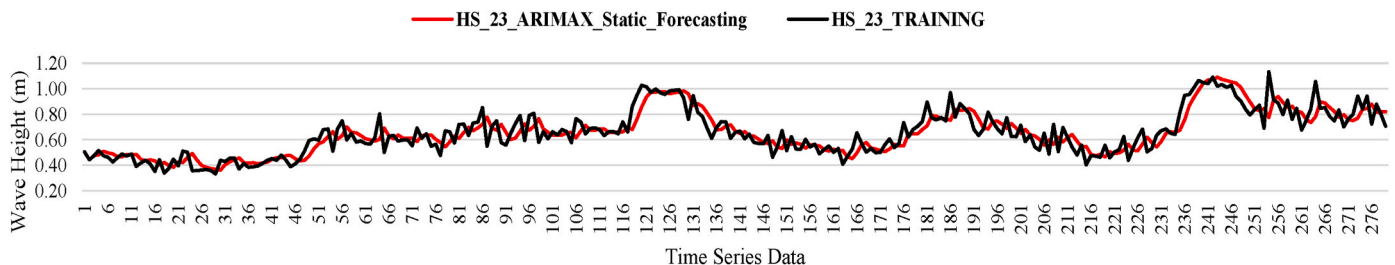


Fig. 10. ARIMAX forecasting of training sample – HS\_23 January.

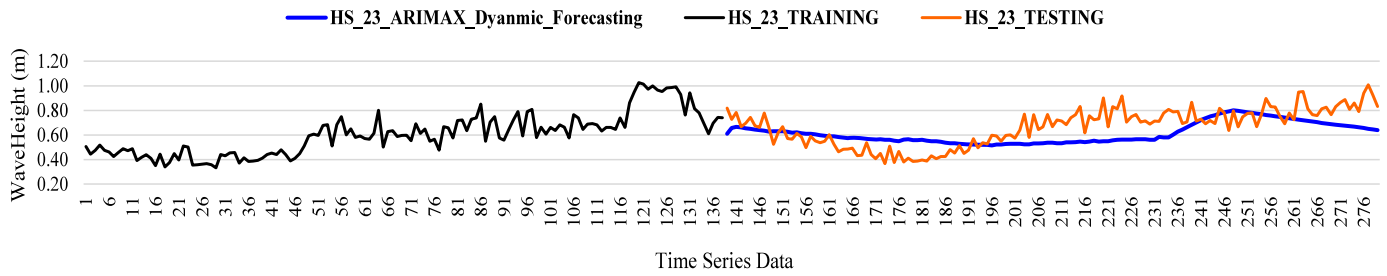


Fig. 11. ARIMAX static, dynamic forecasting, testing sample – January 2023.

close to zero, indicating the model substantially outperforms a naïve benchmark. Although *Theil U2* (2.71332) > 1.0 suggests *U2* underperforms the naïve model in predicting changes, this is less critical than *U1* when evaluating trend accuracy. Although *MAPE* (17.7292 %) and *sMAPE* (18.1767 %) highlight moderate percentage errors, they reflect some difficulty in capturing short-term fluctuations. Nonetheless, the *MAE* (0.8986) indicates a stable average forecast deviation. Overall, the *error metrics* demonstrated that integrating exogenous variables in ARIMAX substantially enhances forecast accuracy, as supported by the values of *U1*, *RMSE*, and *MAE*.

### 8.3. SARIMA

This section analyzes the *Seasonal Autoregressive Integrated Moving Average (SARIMA)* utilizing the *HS\_23* training dataset via *EViews* software. Since unit root testing has already been conducted in previous sections, it will not be repeated here, as we are employing the same candidate model (1,1,2). Thus, the *ADF Correlogram* was used to detect the values for *autoregressive (AR)*, *moving average (MA)*, *seasonal autoregressive (SAR)*, and *seasonal moving average (SMA)*, as shown in Table 8 of the *Pre-Estimation Correlogram*. The diagnosis of this result displays the raw time series data after differencing (based on the *ADF* test). It suggests the *AR* value as *AR(2)*, and the *MA* value is *MA(1)*. The possible lag options of *SAR* and *SMA* are 12, 24, or 36, which do not reveal seasonality. Observing the *AR* values reveals several minor and iterative spikes, making it challenging to assess the presence of seasonality. The *AR* values show minor spikes and display no clear seasonality. Despite this, the estimated *SARIMA* forecasting equation  $d(hs_{23}jfh_{training}) c ar(1) ma(2) sar(12) sma(12)$  was derived with acceptable Coefficient Covariance and low *p-values* < 0.05 for *AR* and *MA*. Overall, *SARIMA Correlogram* does not show significant spikes at lag 12, 24, and 36. It is important to note that, depending on the quality of time series data used for prediction, latent seasonality and structural periodicity may not be clearly visible, resulting in only a partial capture of the underlying trend in the wave structure.

The results in Table 8 of *SARIMA (1,1,1)(1,0,1)(12)* model show statistically significant coefficients for *AR*, *MA*, *SAR*, and *SMA* (*p-value* < 0.05), indicating 23 % of the variance (*R-squared* = 0.23) with strong overall significance (*F-statistic* *p* = 0.0000). Also, diagnostic criteria like *Durbin-Watson stat* (2.0429) suggest a reasonable model fit.

The result in Table 9 exhibits the post-estimation Correlogram, indicating the residuals are white noise after fitting the *SARIMA* model. This ensures that the model has sufficiently captured the data structure prior to forecasting implementation.

The Seasonal *SARIMA* selected model was tested on the hourly wave height data *Hs\_23* with first-order differencing applied for stationarity *I*

Table 8

Pre-estimation correlogram Q-Statistics.

| Autocorrelation | Partial Correlation | AC | PAC    | Q-Stat | Prob         |
|-----------------|---------------------|----|--------|--------|--------------|
| 0.021           | -0.001              | 1  | -0.441 | -0.441 | 54.550 0.000 |
| 0.021           | -0.001              | 2  | 0.022  | -0.213 | 54.691 0.000 |
| 0.041           | -0.001              | 3  | 0.041  | -0.053 | 55.169 0.000 |
| -0.001          | 0.005               | 4  | -0.001 | 0.005  | 55.169 0.000 |
| 0.041           | 0.008               | 5  | 0.048  | 0.080  | 55.833 0.000 |
| -0.001          | 0.001               | 6  | -0.001 | 0.001  | 56.889 0.000 |
| -0.001          | -0.010              | 7  | -0.058 | -0.109 | 57.863 0.000 |
| 0.021           | 0.007               | 8  | 0.122  | 0.037  | 62.126 0.000 |
| -0.001          | 0.001               | 9  | -0.062 | 0.010  | 63.230 0.000 |
| 0.070           | 0.003               | 10 | 0.070  | 0.093  | 64.671 0.000 |
| -0.001          | -0.001              | 11 | -0.084 | -0.017 | 66.736 0.000 |
| 0.070           | -0.003              | 12 | 0.030  | -0.019 | 67.005 0.000 |
| 0.011           | -0.002              | 13 | 0.010  | -0.022 | 67.035 0.000 |
| -0.001          | -0.001              | 14 | -0.049 | -0.058 | 67.752 0.000 |
| 0.070           | 0.003               | 15 | 0.074  | 0.053  | 69.375 0.000 |
| -0.001          | -0.001              | 16 | -0.119 | -0.083 | 73.558 0.000 |
| 0.070           | -0.003              | 17 | 0.071  | -0.013 | 75.048 0.000 |
| -0.001          | -0.001              | 18 | -0.029 | -0.050 | 75.298 0.000 |
| -0.001          | -0.001              | 19 | -0.014 | -0.030 | 75.360 0.000 |
| 0.021           | -0.008              | 20 | 0.021  | -0.008 | 75.494 0.000 |
| -0.001          | -0.001              | 21 | -0.087 | -0.092 | 77.799 0.000 |
| 0.041           | 0.006               | 22 | 0.148  | 0.096  | 84.427 0.000 |
| -0.001          | -0.001              | 23 | -0.123 | -0.055 | 89.056 0.000 |
| -0.001          | -0.001              | 24 | -0.031 | -0.087 | 89.355 0.000 |
| 0.070           | -0.003              | 25 | 0.079  | -0.018 | 91.285 0.000 |
| -0.001          | 0.002               | 26 | -0.044 | 0.002  | 91.893 0.000 |
| -0.001          | -0.001              | 27 | -0.049 | -0.092 | 92.641 0.000 |
| -0.001          | -0.001              | 28 | -0.018 | -0.107 | 92.738 0.000 |
| 0.070           | 0.005               | 29 | 0.078  | 0.055  | 94.645 0.000 |
| 0.021           | 0.007               | 30 | 0.026  | 0.057  | 94.849 0.000 |
| -0.001          | -0.003              | 31 | -0.088 | -0.003 | 97.285 0.000 |
| -0.001          | -0.001              | 32 | -0.008 | -0.092 | 97.303 0.000 |
| 0.070           | 0.001               | 33 | 0.070  | 0.021  | 98.845 0.000 |
| -0.001          | -0.004              | 34 | -0.024 | -0.004 | 99.037 0.000 |
| -0.001          | -0.002              | 35 | -0.011 | -0.012 | 99.076 0.000 |
| -0.001          | -0.003              | 36 | -0.032 | -0.005 | 99.399 0.000 |

(1). The seasonal lag is based on a 12-period cycle, proposing that the daily wave heights are increasing and consistent with the potential patterns of the wave significant heights during the first half of January 2023. The resulting linear trend reflects the internal logic and nature of dynamic forecasting over an extended period, as depicted in Fig. 12. Concluding that the model prioritizes capturing moderate noise while avoiding replicating high-frequency seasonal oscillations.

#### 8.3.1. SARIMA forecast evaluation

The *SARIMA* forecast model was evaluated using error metrics that reflect its relatively weaker forecasting performance. The model produced an *RMSE* (0.2837), reflecting an average forecast error. *Theil U1* statistic (0.1852) is relatively low, indicating a reasonable error level. *Theil U2* (9.0170) > 1.0 suggests *U2* is very high and worse than the naïve model in predicting changes. *MAPE* (47.5761 %) and *sMAPE* (34.4659 %) both highlight high percentage errors and difficulty capturing short-term fluctuations. Nonetheless, the *MAE* (0.2422) confirms an average absolute error, as depicted in Table 10. Collectively, the

Table 7

ARIMAX forecast evaluation for *HS\_23*.

| Forecast Evaluation Statistics |        |        |          |           |          |          |        |
|--------------------------------|--------|--------|----------|-----------|----------|----------|--------|
| ARIMAX Forecast                | RMSE   | MAE    | MAPE (%) | sMAPE (%) | Theil U1 | Theil U2 | RAE    |
| HS_23_Forecasting              | 0.1403 | 0.1148 | 17.7292  | 18.1767   | 0.1076   | 2.71332  | 0.8986 |

**Table 9**  
Post-estimation correlogram Q-Statistics.

| Autocorrelation | Partial Correlation | AC     | PAC    | Q-Stat | Prob  |
|-----------------|---------------------|--------|--------|--------|-------|
| 1               | 0.000               | -0.027 | -0.027 | 0.006  |       |
| 2               | 0.000               | -0.005 | -0.006 | 0.208  |       |
| 3               | 0.000               | 0.081  | 0.081  | 2.087  |       |
| 4               | 0.000               | 0.033  | 0.037  | 2.389  |       |
| 5               | 0.000               | 0.043  | 0.046  | 2.919  | 0.088 |
| 6               | 0.000               | -0.079 | -0.040 | 4.701  | 0.095 |
| 7               | 0.000               | -0.038 | -0.049 | 5.121  | 0.163 |
| 8               | 0.000               | 0.108  | 0.098  | 8.484  | 0.075 |
| 9               | 0.000               | 0.003  | 0.020  | 8.487  | 0.131 |
| 10              | 0.000               | 0.041  | 0.054  | 8.974  | 0.175 |
| 11              | 0.000               | -0.080 | -0.088 | 10.829 | 0.146 |
| 12              | 0.000               | 0.036  | 0.019  | 11.214 | 0.190 |
| 13              | 0.000               | -0.025 | -0.049 | 11.397 | 0.249 |
| 14              | 0.000               | -0.042 | -0.018 | 11.926 | 0.290 |
| 15              | 0.000               | 0.004  | 0.011  | 11.930 | 0.369 |
| 16              | 0.000               | -0.107 | -0.103 | 15.327 | 0.224 |
| 17              | 0.000               | 0.014  | 0.004  | 15.381 | 0.284 |
| 18              | 0.000               | -0.044 | -0.057 | 15.957 | 0.316 |
| 19              | 0.000               | -0.047 | -0.015 | 16.614 | 0.342 |
| 20              | 0.000               | -0.023 | -0.035 | 16.769 | 0.401 |
| 21              | 0.000               | -0.068 | -0.045 | 18.167 | 0.378 |
| 22              | 0.000               | 0.079  | 0.071  | 20.051 | 0.330 |
| 23              | 0.000               | -0.137 | -0.136 | 25.814 | 0.135 |
| 24              | 0.000               | -0.043 | -0.020 | 26.380 | 0.154 |
| 25              | 0.000               | 0.024  | -0.002 | 26.551 | 0.186 |
| 26              | 0.000               | -0.063 | -0.029 | 27.759 | 0.184 |
| 27              | 0.000               | -0.099 | -0.120 | 30.771 | 0.129 |
| 28              | 0.000               | -0.021 | 0.001  | 30.914 | 0.156 |
| 29              | 0.000               | 0.088  | 0.094  | 33.337 | 0.123 |
| 30              | 0.000               | 0.012  | -0.008 | 33.385 | 0.151 |
| 31              | 0.000               | -0.103 | -0.070 | 36.748 | 0.100 |
| 32              | 0.000               | -0.023 | -0.070 | 36.913 | 0.121 |
| 33              | 0.000               | 0.051  | 0.052  | 37.757 | 0.128 |
| 34              | 0.000               | -0.001 | -0.031 | 37.757 | 0.156 |
| 35              | 0.000               | -0.051 | -0.010 | 38.581 | 0.164 |
| 36              | 0.000               | -0.002 | 0.008  | 38.582 | 0.196 |

error metrics suggest that *SARIMA* model performs weaker than the *ARIMAX* forecasting performance.

#### 8.4. Vector autoregressive (VAR) model

The main use of VAR models for forecasting is to enable the analysis of multivariate time series data by generalizing several univariate autoregressive models. The difference between univariate models, such as *ARIMA*, and multivariate models, such as VAR, and their lagged values clarifies present values. In this instance, a VAR model consists of  $n$  variables, thus defining an  $n$ -equation model. Each VAR single variable is represented as a linear function of its past values, which is the same scenario as other variables, where their serial error terms are uncorrelated. The observed discrepancy between actual and predicted wave static and dynamics with *ARIMA*, *ARIMAX*, and *SARIMA* models suggests further refinement is necessary. Exploring the influence of multivariate variables in models such as the VAR model is also possible. Utilizing the VAR model offers a comprehensive frequentist set in the multivariate analysis of time series data. The expectations of VAR

models derive predictive routines. The following equations (39)–(41) denote the general formula of the VAR model based on the lag order selection criteria and constant terms defined as  $C$ .

Multivariate VAR( $n$ ) model ( $x_t, y_t, z_t$ ):

$$x_t = a_1 + \sum_{k=1}^{k=4} b_{11}x_{t-k} + \sum_{m=1}^{m=4} b_{12}y_{t-m} + \sum_{p=1}^{p=4} b_{13}z_{t-p} + u_t \quad (39)$$

$$y_t = a_2 + \sum_{k=1}^{k=4} b_{21}x_{t-k} + \sum_{m=1}^{m=4} b_{22}y_{t-m} + \sum_{p=1}^{p=4} b_{23}z_{t-p} + v_t \quad (40)$$

$$z_t = a_3 + \sum_{k=1}^{k=4} b_{31}x_{t-k} + \sum_{m=1}^{m=4} b_{32}y_{t-m} + \sum_{p=1}^{p=4} b_{33}z_{t-p} + w_t \quad (41)$$

$x_t, y_t, z_t$ : Stationary variables (Logs and Differences could be applied)

$x_{t-1}, y_{t-1}, z_{t-1}$ : Lagged variables (Past variables)

$u_t, v_t, w_t$ : White noise (Shock terms of disturbances)

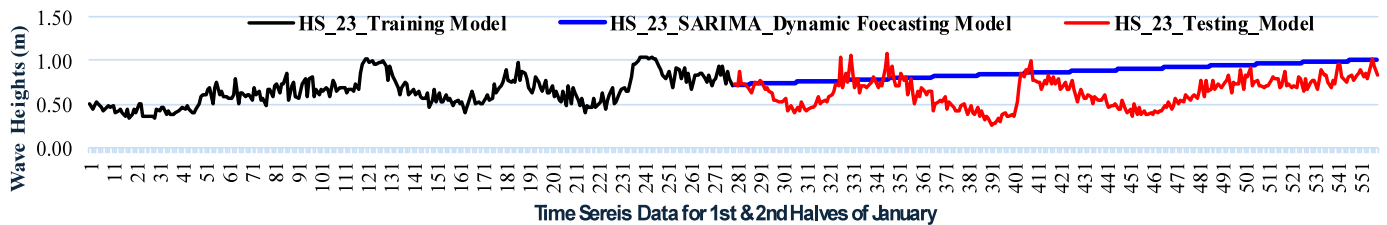
$a_1, a_2, a_3$ : Constants

$b_{11}, b_{12}, b_{21}, b_{22}, b_{31}, b_{32}$ : Lagged values of  $x_t, y_t, z_t$

##### 8.4.1. VAR estimates and lag order selection

Three endogenous variables were selected for the VAR model estimate: (1) Significant Wave Height ( $H_{23}$ ), (2) Wind Speed ( $WSPD_{23}$ ), and (3) Dominant Wave Period ( $DPD_{23}$ ). Also, a constant  $C$  was selected as an exogenous variable. An appropriate lag length is an essential factor, where a too small lag length may mis-specify the model, and a too large lag length causes waste in degrees of freedom. In comparison, the frequency of the data supports defining the lag length. For instance, annual data is equivalent to one lag, monthly data to 12 lags, and quarterly data to 4 lags. In our case, 12 lags might be the option; however, using lag length criteria with the condition of no autocorrelation at the selected lag is also an option where four criteria used in this regard: (1) Likelihood Ratio Test statistics, (2) Akaike Information Criteria (AIC), (3) Schwartz Criterion (SC), (4) Hannan-Quinn Criterion (H-Q), (5) Log Likelihood (LR), (6) Final Prediction Error (FPE). Based on the above discussion, we have examined the lag structure utilizing lag length criteria by selecting 12 lags as shown in Table 11, with a preference for lags 3 and 11 as denoted by asterisks.

Referring to the results in Table 11, Lag 3 was selected. The VAR stability condition is required to define the model's stationarity condition and residual diagnostics. Fig. 13 indicates the VAR stability, which can be achieved when the inverse roots of the AR polynomial contain a modulus  $<1.0$  and lie within the unit circle. The inverse roots of the AR characteristics of the Polynomial plot, denoted by the black dots, lie



**Fig. 12.** *SARIMA* Dynamic Forecasting Model combined with Training and Testing Samples.

**Table 10**  
*SARIMA* forecast evaluation for  $HS_{23}$ .

| Forecast Evaluation Statistics |        |        |          |           |          |          |        |
|--------------------------------|--------|--------|----------|-----------|----------|----------|--------|
| <i>SARIMA</i> Forecast         | RMSE   | MAE    | MAPE (%) | sMAPE (%) | Theil U1 | Theil U2 | RAE    |
| $HS_{23}$ Forecasting          | 0.2837 | 0.2422 | 47.5761  | 34.4659   | 0.1852   | 9.0170   | 1.7087 |

**Table 11**  
Lag length criteria – VAR Model.

| Lag | LogL      | LR        | FPE       | AIC       | SC        | HQ        |
|-----|-----------|-----------|-----------|-----------|-----------|-----------|
| 0   | -827.3432 | NA        | 0.100879  | 6.219799  | 6.260105  | 6.235990  |
| 1   | -387.9886 | 865.5450  | 0.004016  | 2.996169  | 3.157394  | 3.060932  |
| 2   | -348.5015 | 76.90362  | 0.003196  | 2.767802  | 3.049945  | 2.881137  |
| 3   | -315.2105 | 64.08830  | 0.002665* | 2.585847* | 2.988908* | 2.747754* |
| 4   | -308.3051 | 13.13842  | 0.002707  | 2.601536  | 3.125516  | 2.812016  |
| 5   | -304.3174 | 7.497410  | 0.002811  | 2.639082  | 3.283980  | 2.898133  |
| 6   | -298.9692 | 9.935277  | 0.002890  | 2.666436  | 3.432253  | 2.974060  |
| 7   | -294.2833 | 8.599609  | 0.002986  | 2.698751  | 3.585487  | 3.054947  |
| 8   | -285.2716 | 16.33581  | 0.002988  | 2.698664  | 3.706318  | 3.103431  |
| 9   | -283.2039 | 3.701670  | 0.003149  | 2.750591  | 3.879164  | 3.203931  |
| 10  | -281.8361 | 2.418084  | 0.003337  | 2.807761  | 4.057252  | 3.309673  |
| 11  | -271.8899 | 17.35912* | 0.003317  | 2.800674  | 4.171083  | 3.351158  |
| 12  | -264.6026 | 12.55504  | 0.003364  | 2.813502  | 4.304830  | 3.412559  |

inside the blue unit circle, indicating the VAR model satisfies the stability condition, suitable for forecasting. This clarifies that the VAR model is stationary and rejects nonstationary problems. Based on this, the VAR model satisfies the stability condition.

#### 8.4.2. VAR lag 3 of serial correlation LM tests

Before proceeding further in VAR analysis, the adequacy of the model should be defined, and the estimate of the VAR model should not be white noise, while errors from the true VAR are white noise. This test is based on multivariate portmanteau statistics to assess the presence of autocorrelation in the residuals of a *vector autoregressive* (VAR) model, and is defined as  $Q_h$  statistic. This is to test the null hypothesis of no autocorrelation up to the lag  $h$  in the residuals of the VAR model as depicted in Equations (42) and (43) (Reggiannini et al., 2024). These checks are necessary to ensure the model's stationarity condition for the validity of the VAR model.

$$Q_h = T \sum_{j=1}^h \text{tr}(\hat{C}_j \hat{C}_0^{-1} \hat{C}_j \hat{C}_0^{-1}) \quad (42)$$

$$\hat{C}_j = T^{-1} \sum_{t=j+1}^T \hat{u}_t \hat{u}_{t-j}' \quad (43)$$

$T$ : the sample size,

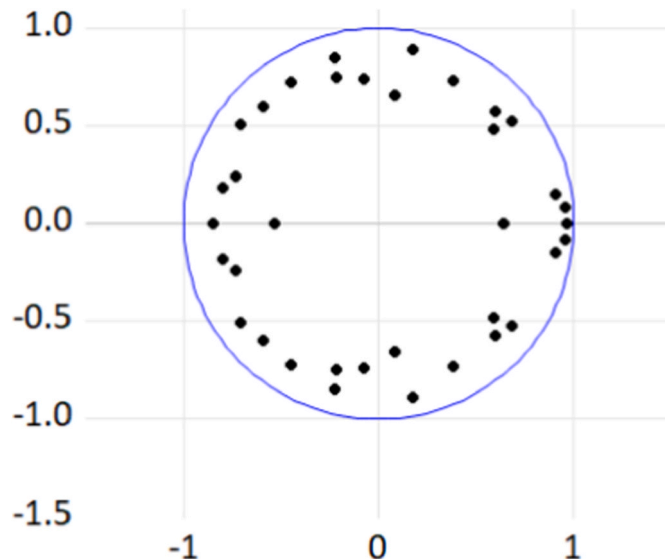
$h$ : the number of lags being tested,

$\hat{C}_j$ : estimates the autocovariance matrix of the residual at lag  $j$ ,

$\hat{C}_0$ : estimates the autocovariance matrix of the residual at lag 0,

$\text{tr}(\cdot)$ : the trace operator, the sum of the diagonal elements of a matrix

$\hat{u}_t$ : estimated Residuals



**Fig. 13.** Inverse roots of AR characteristics.

This analysis also aims to clarify the spatiotemporal interdependence relationships between the three variables to ensure that the time series data has been transformed into stationarity. This is a critical requirement for time series modelling via VAR. Accordingly, selecting an appropriate lag length with no autocorrelation is essential to avoid the model's misspecifications or wasting its degrees of freedom. Referring to Table 12, the autocorrelation results of the tested Lag 3 show evidence of  $p$ -values  $> 0.05$ , indicating that those residuals are free from serial correlation and long enough to capture data dynamics. The VAR residual test of the autocorrelation Lag 3 successfully rejected the null hypothesis, where all  $p$ -values  $> 0.05$ .

We chose to apply *Correlograms* via the *Residuals Test* to define the  $p$ -value, where the ideal scenario is when the  $p$ -value  $> 0.05$  indicates non-significant serial correlation at certain lags. This also means the VAR model has adequately captured the data's relationships and dynamics. The residual values (errors) behave randomly, as it is the condition required for the model to be nominated for forecasting and interpretation. Oppositely, when a  $p$ -value  $< 0.05$  is proposed, it implies that some essential dynamics are missing in the model, and further attention is required to adjust the model, such as adding more lags or modifying the model structure.

#### 8.4.3. VAR static forecasting of training model

The VAR (*Vector Autoregression*) model captures the relationships between multivariate time series by using a linear function based on recent data lags for each variable at a given point in time. This model can be represented in matrix form, as illustrated in regression Equation (44), or in scalar notation, as outlined in Equations (45)–(47). This explains its use in analyzing time series data (Ichsan et al., 2019), (Tu et al., 2018).

$$\begin{bmatrix} Y_{HS\_21,t} \\ Y_{HS\_22,t} \\ Y_{HS\_23,t} \end{bmatrix} = \begin{bmatrix} C_{HS\_21} \\ C_{HS\_22} \\ C_{HS\_23} \end{bmatrix} + \begin{bmatrix} A_{1,1} & A_{1,2} & A_{1,3} \\ A_{2,1} & A_{2,2} & A_{2,3} \\ A_{3,1} & A_{3,2} & A_{3,3} \end{bmatrix} \begin{bmatrix} Y_{HS\_21,t-1} \\ Y_{HS\_22,t-1} \\ Y_{HS\_23,t-1} \end{bmatrix} + \begin{bmatrix} e_{HS\_21,t} \\ e_{HS\_22,t} \\ e_{HS\_23,t} \end{bmatrix} \quad (44)$$

$$Y_{HS\_21,t} = C_{HS\_21} + A_{1,1}Y_{HS\_21,t-1} + A_{1,2}Y_{HS\_22,t-1} + A_{1,3}Y_{HS\_23,t-1} + e_{HS\_21,t} \quad (45)$$

$$Y_{HS\_22,t} = C_{HS\_22} + A_{2,1}Y_{HS\_21,t-1} + A_{2,2}Y_{HS\_22,t-1} + A_{2,3}Y_{HS\_23,t-1} + e_{HS\_22,t} \quad (46)$$

$$Y_{HS\_23,t} = C_{HS\_23} + A_{3,1}Y_{HS\_21,t-1} + A_{3,2}Y_{HS\_22,t-1} + A_{3,3}Y_{HS\_23,t-1} + e_{HS\_23,t} \quad (47)$$

Fig. 14 displays the VAR model results to compare the actual and static forecasted values for  $HS_{23}$ . The forecast results closely align with the actual values and capture the occasions of various fluctuations well. Some minor deviations were observed in regions of high variability, but overall, the model-fitting predictions align very closely with the actual data.

**Table 12**  
VAR residual serial correlation LM tests.

| Null hypothesis: No serial correlation at lag h       |           |    |        |            |             |        |
|-------------------------------------------------------|-----------|----|--------|------------|-------------|--------|
| Lag                                                   | LRE* stat | df | Prob.  | Rao F-stat | df          | Prob.  |
| 1                                                     | 7.879693  | 9  | 0.5463 | 0.876210   | (9, 547.7)  | 0.5463 |
| 2                                                     | 13.63926  | 9  | 0.1357 | 1.524636   | (9, 547.7)  | 0.1358 |
| 3                                                     | 12.51714  | 9  | 0.1857 | 1.397772   | (9, 547.7)  | 0.1857 |
| Null hypothesis: No serial correlation at lags 1 to h |           |    |        |            |             |        |
| Lag                                                   | LRE* stat | df | Prob.  | Rao F-stat | df          | Prob.  |
| 1                                                     | 7.879693  | 9  | 0.5463 | 0.876210   | (9, 547.7)  | 0.5463 |
| 2                                                     | 23.20520  | 18 | 0.1828 | 1.296465   | (18, 628.4) | 0.1829 |
| 3                                                     | 37.49386  | 27 | 0.0862 | 1.401948   | (27, 640.2) | 0.0863 |

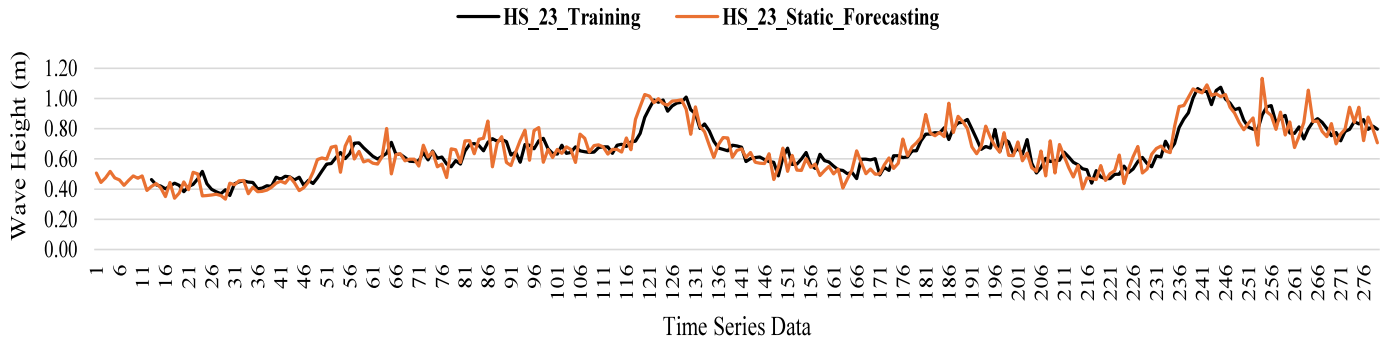


Fig. 14. VAR Static forecasting of training sample HS\_23\_January.

#### 8.4.4. VAR dynamic forecasting of testing model

VAR dynamic forecasting is an extension of an autoregressive ( $p$ ) model to predict future values, as outlined in Equations (48)–(53). In conjunction with the VAR forecasting, the model suggests bias upon stationarity but over-reliance on past averages rather than trend adaptations.

$$Y_t = \mu + \varepsilon_t + \psi_{\varepsilon_{t-1}} + \psi_{2\varepsilon_{t-2}} + \dots \quad (48)$$

$$Y_t = \mu + \varepsilon_t + \psi_{(1 \varepsilon_{t+h-1})} + \dots + \psi_{(h-1 \varepsilon_{t+1})} \dots \quad (49)$$

$$E(Y_t) = \mu \quad (50)$$

$$VAR(Y_t) = E(Y_t - \mu)(Y_t - \mu) \quad (51)$$

$$VAR(Y_t) = E \left[ \left( \sum_{k=1}^{\infty} \psi_{k\varepsilon_{t-k}} \right) \left( \sum_{k=1}^{\infty} \psi_{k\varepsilon_{t-k}} \right)' \right] \quad (52)$$

$$VAR(Y_t) = \sum_{k=1}^{\infty} \psi_k \sum \psi_k' \quad (53)$$

Fig. 15 compares the VAR dynamic forecast (blue line) with the testing sample (orange line) for HS\_23 (wave height). The dynamic forecast captures the general trend of the wave height; nonetheless, the VAR model prioritizes capturing moderate noise while avoiding replicating high-frequency seasonal oscillations.

#### 8.4.5. VAR forecast evaluation

Table 13 displays the VAR forecast evaluation using error metrics that indicate a moderate forecasting performance compared to the other models. The VAR model results achieved an RMSE (0.1739), reflecting a low forecast error. Theil U1 statistic (0.1284) is relatively low, indicating a better accuracy than ARIMA and SARIMA, but not surpassing ARIMAX. Theil U2 (3.3900) > 1.0 suggests U2 is high and worse than the naïve model in predicting changes. MAPE (27.5033 %) and sMAPE (22.6957 %) both indicate moderate percentage errors that challenge the model's

ability to capture short-term fluctuations. Nonetheless, the MAE (0.9899) confirms an average absolute error. Overall, despite the forecasting curve result in Fig. 16, several evaluation parameters in Table 13 demonstrate that the VAR model exhibits reasonable forecasting performance.

The broader purpose of using ARIMA, SARIMA, ARIMAX, and VAR models is to predict significant wave heights at specific spatiotemporal locations to facilitate effective ship routing and decision making. The goal is to provide a reliable forecast for a period of at least two weeks to one month, aiding captains and ship owners in planning their ship routes efficiently. However, with these models, shorter-term dynamic forecasts have proven to be more effective in capturing trends and fluctuations, as demonstrated in Figs. 16–18. This limitation does not fully align with the practical operational needs of ship routing plans. Fig. 17 displays the effectiveness of dynamic and static forecasts based on 40 data points, indicating successful prediction. Also, dynamic predictions illustrated in Fig. 17, utilizing the same sample of time series data we have used in this study, show an upward prediction where the wave reaches 1.2 m in height. In contrast, the subsequent forecasts in Fig. 18, derived from the same dataset but extended by 40 additional points, demonstrate downward fluctuations. Despite accurate forecasting provided by the VAR model, the inherent limitations of short-term forecasting with the ship routing goal require a broader prediction of wave heights over a longer period. Overall, shorter prediction windows enable models such as VAR and other models to align better with actual data variability, although their reliability diminishes over longer periods. Given these observations, optimizing the cyclic wave height prediction windows based on operational needs may require exploring hybrid modeling approaches such as combining ARIMA, ARIMAX, SARIMA, and VAR with machine learning techniques to better capture long-term volatility and trends.

## 9. Data reliability

This study relies on NDBC buoy data, which undergo strict quality assurance procedures, including NIST-traceable calibration, WMO GTS

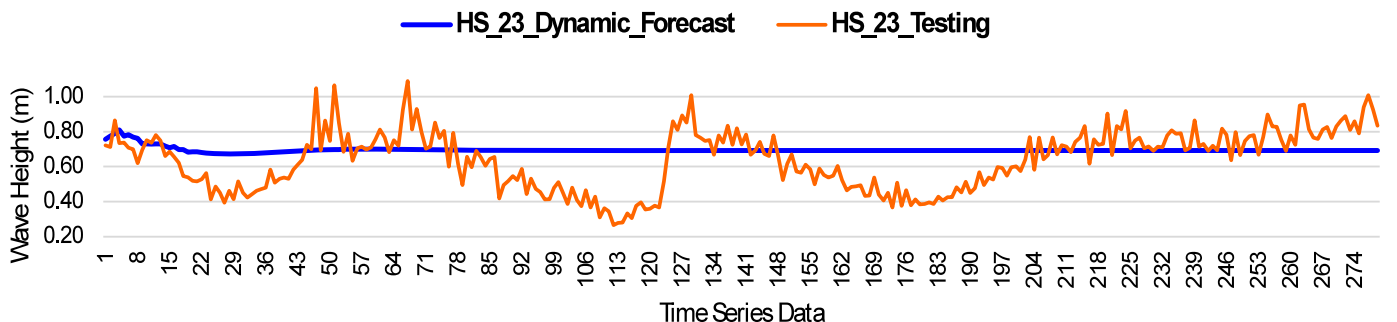


Fig. 15. VAR dynamic forecasting Vs. Testing sample.

**Table 13**  
VAR forecast evaluation for *HS\_23*.

| Forecast Evaluation Statistics |        |        |          |           |          |          |        |
|--------------------------------|--------|--------|----------|-----------|----------|----------|--------|
| VAR Forecast                   | RMSE   | MAE    | MAPE (%) | sMAPE (%) | Theil U1 | Theil U2 | RAE    |
| HS_23_Forecasting              | 0.1739 | 0.1403 | 27.5033  | 22.6957   | 0.1284   | 3.3900   | 0.9899 |

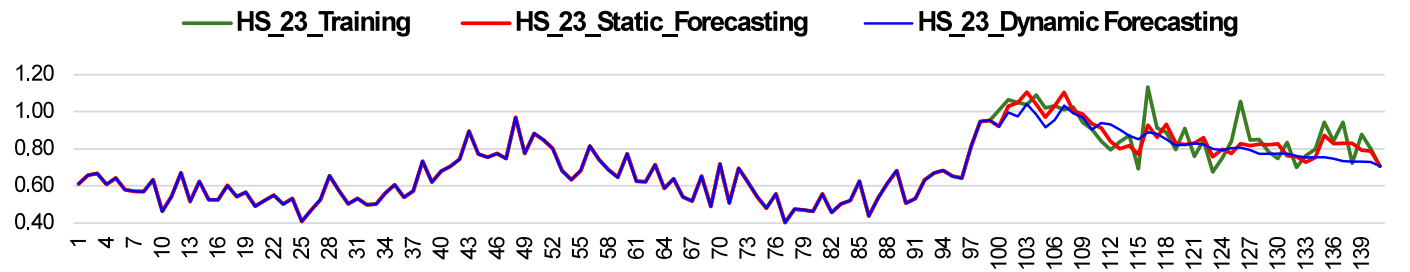


Fig. 16. VAR static & dynamic forecasting Vs. Training sample.

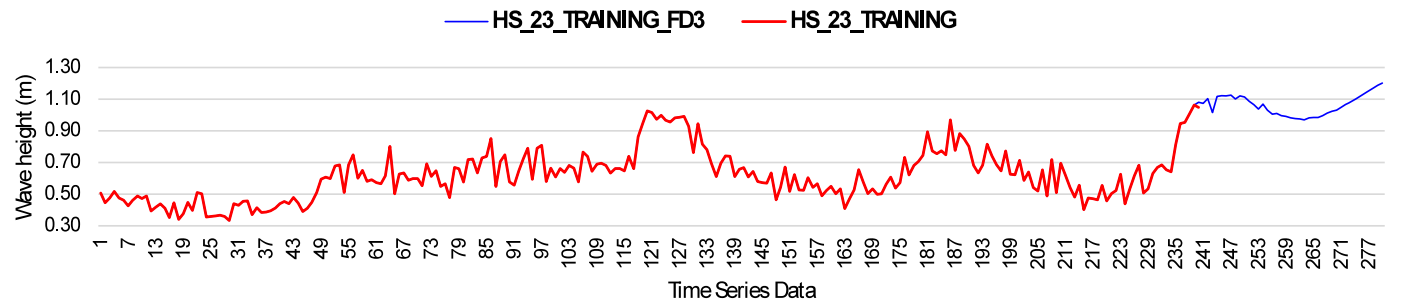


Fig. 17. VAR dynamic forecasting from data point 240 to data point 280.

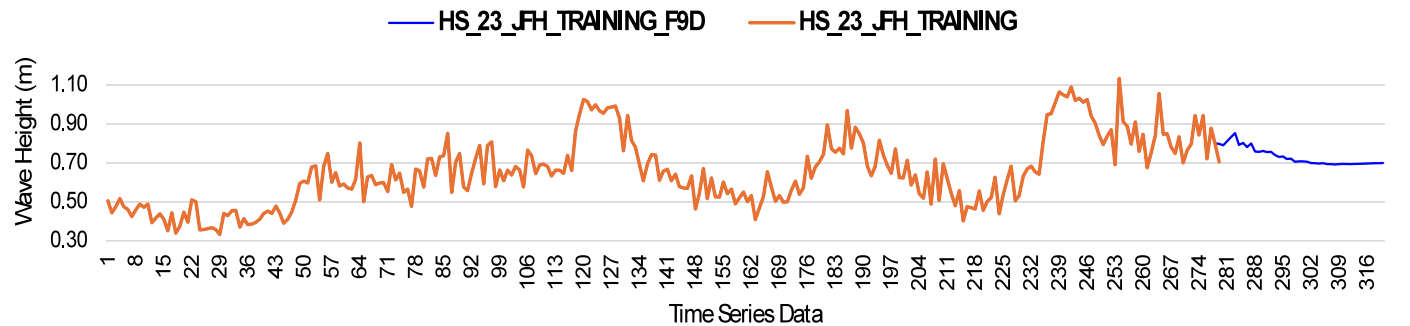


Fig. 18. VAR dynamic forecasting from data point 280 to data point 320.

formatting standards, QARTOD automated QC checks, and cross-validation with NCEP forecast models. These institutional protocols ensure measures of accuracy and consistency. Additional preprocessing steps, outlier filtering, interpolation of minor gaps, and stationarity testing further strengthen the dataset's reliability and support reproducible forecasting results.

## 10. Limitations

The forecasting models used ARIMA, SARIMA, ARIMAX, and VAR are limited by their analytical structure to capture rapid nonlinear wave fluctuations. The analysis is based on a single buoy station, reducing spatial representativeness for vessels moving across different sea states. The study does not include ship-specific resistance or engine models. Only a limited set of environmental variables was incorporated; additional predictors may further improve performance, but were outside this study's scope.

## 11. Discussion

The evaluation of the four classical time series models (ARIMA, ARIMAX, SARIMA, and VAR) enables direct assessment of the research hypotheses. The null hypothesis (*H0*) proposed that classical models would not significantly capture trends or fluctuations in wave heights and, therefore, would not enhance the Decision Support System (DSS). This hypothesis is rejected, as all models demonstrated some level of predictive capability, particularly in representing long-term trends. Hypothesis one (*H1*) stated that classical models improve prediction by capturing only long-term trends, thereby partially enhancing DSS performance. The results support this hypothesis. ARIMA and SARIMA performed reasonably well in trend detection, as reflected in their RMSE and MAE values, but both exhibited high percentage errors (MAPE >46 %), confirming that their primary strength lies in representing overall wave height trends rather than short-term fluctuations. Similarly, the VAR model achieved moderate accuracy, with MAPE of 27.50 % and

Theil U1 of 0.1284, further supporting H1 by capturing long-term relationships through multivariate analysis.

Hypothesis 2 (H2) posits that classical models would improve prediction by capturing both short-term fluctuations and long-term trends, thus fully enhancing DSS performance. This hypothesis is not supported. None of the models consistently reproduced short-term variability with sufficient accuracy. Even though ARIMAX clearly outperformed the other models, with RMSE = 0.1403, MAE = 0.1148, and MAPE = 17.73 %, it still failed to dynamically model short-term fluctuations across the testing dataset. This limitation indicates that while exogenous variables improve predictive accuracy, they are not sufficient to achieve full dual-scale (trend and fluctuation) forecasting with a DSS context. Overall, the findings demonstrate that Hypothesis 1 is supported while Hypothesis 2 is not. The models contribute partially to DSS integration by reliably capturing trends, but their inability to reproduce short-term fluctuations highlights the need for more advanced or hybrid forecasting approaches to enhance ship routing decisions in highly dynamic sea states.

Regarding the study limitations, this study focuses exclusively on classical time series prediction models, specifically ARIMA, SARIMA, ARIMAX, and VAR, utilizing EViews software. These classical approaches offer deterministic predictions based on observed data patterns, making them a valuable tool for straightforward forecasting. However, their predictive accuracy is sensitive to the length and quality of the input data points. To further our research goal, Machine Learning (ML) approaches offer substantial advantages in dealing with a variety of complex, nonlinear, and diverse datasets, whether short, long, or sparse. These future targeted ML methods leverage prior distributions and make predictions based on uncertainty. This adaptability enables the smooth integration of prior beliefs and external information into the forecasting process. Consequently, there is a valuable opportunity to transition from classical model applications to the framework of machine learning tools or to hybridize them, which could lead to meaningful advancements in analysis and enhance the model's predictive power.

## 12. Conclusion

The choice of univariate and multivariate classical models in this study was reasonable and aligned with the time series data of variables and research objectives. When predicting ship routing based on significant wave height, it is clear that trend predictions are often more valuable than precise fluctuations and certain point estimates. This study applied four analytical models: ARIMA, ARIMAX, SARIMA, and VAR to forecast significant wave height (HS\_23) for January 2023, evaluating both static and dynamic forecasting performance. The aim of this study is to focus on the utility of ship route modeling in maritime engineering research aimed at developing a robust decision support system for ship route forecasting. Utilizing spatiotemporal univariate and multivariate time series parameters based on three key variables: HS\_23, DPD\_23, and WSPD\_23. The methodology harnesses autoregressions with moving averages, and vector autoregression with hierarchical prior selection to deliver a precise and transparent assessment

**Table 14**  
ARIMA, SARIMA, ARIMAX, VAR forecast evaluation for HS\_23.

| Forecast Evaluation Statistics |        |        |          |           |          |          |
|--------------------------------|--------|--------|----------|-----------|----------|----------|
|                                | RMSE   | MAE    | MAPE (%) | SMAPE (%) | Theil U1 | Theil U2 |
| ARIMA Forecast                 | 0.2798 | 0.2382 | 46.9037  | 34.0510   | 0.1833   | 8.77430  |
| ARIMAX Forecast                | 0.1403 | 0.1148 | 17.7292  | 18.1767   | 0.1076   | 2.71332  |
| SARIMA Forecast                | 0.2837 | 0.2422 | 47.5761  | 34.4659   | 0.1852   | 9.01700  |
| VAR Forecast                   | 0.1739 | 0.1403 | 27.5033  | 22.6957   | 0.1284   | 3.39000  |

of significant wave heights. Static forecasting across all models showed strong alignment with the training time series data, effectively capturing both trends and fluctuations within the dataset. The static forecasting results confirm that the models accurately replicate historical patterns within the confines of the available time series data. However, the primary objective of this study is dynamic forecasting. Specifically, predicting future wave heights beyond the current dataset. This goal revealed significant differences among the models used. Based on the results, while the null hypothesis was rejected, only Hypothesis 1 was supported, whereas Hypothesis 2 was not. The findings demonstrate that most of the classical time series models capture long-term trends but are limited in representing short-term fluctuations. Given that the primary aim of this study is to forecast significant wave heights, the ability to model trend behaviour is of practical relevance, making the prediction suitable for enhancing DSS in ship routing applications. The conclusion of the forecast evaluation in Table 14 demonstrates that the ARIMAX model outperforms other models across most metrics, achieving the lowest values of RMSE (0.1403), MAE (0.1148), and MAPE (17.73 %), indicating higher predictive accuracy and lower error magnitudes. In contrast, ARIMA and SARIMA show higher error levels, particularly in percentage-based metrics (MAPE > 46 %). Finally, the VAR model performs moderately well, especially in MAPE (27.5 %) and Theil U1 (0.1284), though it does not surpass ARIMAX. These results confirm that integrating exogenous variables in ARIMAX improves accuracy over other time series models. However, it has not yet fully succeeded in

**Table 15**  
The nomenclature used in this study.

| S. No. | Parameter Notation     | Parameter Definition                                       | Unit                            |
|--------|------------------------|------------------------------------------------------------|---------------------------------|
| 1      | $V_w$                  | Wave Velocity                                              | m/s                             |
| 2      | $L_w$                  | Wavelength                                                 | m                               |
| 3      | $T_w$                  | Wave Period                                                | s                               |
| 4      | $P$                    | Pressure distribution under a wave                         | Pa (N/m <sup>2</sup> )          |
| 5      | $k$                    | Wave number                                                | Rad/m                           |
| 6      | $H_s$                  | Wave Height                                                | m                               |
| 7      | $m_0$                  | Spectral zeroth moments                                    | m <sup>2</sup>                  |
| 8      | $m_1$                  | Spectral first moments                                     | m <sup>2</sup> . s              |
| 9      | $m_2$                  | Spectral second moments                                    | m <sup>2</sup> . s <sup>2</sup> |
| 10     | $m_3$                  | Spectral third moments                                     | m <sup>2</sup> . s <sup>3</sup> |
| 11     | $m_4$                  | Spectral fourth moments                                    | m <sup>2</sup> . s <sup>4</sup> |
| 12     | $T_0 = T_p$            | Wave periods overall wave climate                          | s                               |
| 13     | $T_1$                  | Mean wave period of the highest wave amplitude             | s                               |
| 14     | $N_0$                  | mean zero-crossing frequency                               | S <sup>-1</sup>                 |
| 15     | $SB\zeta_1$            | Bretschneider spectrum                                     | m <sup>2</sup> /(rad/sec)       |
| 16     | $B_1, B_2$             | Spectrum coefficients                                      |                                 |
| 17     | $SJ\zeta_1, SJ\zeta_2$ | JONSWAP Spectrums                                          | m <sup>2</sup> /(rad/sec)       |
| 18     | $\alpha_M$             | Maximum Wave slope (first approximation)                   | dimensionless                   |
| 19     | $E$                    | Wave Energy per unit area                                  | J/m <sup>2</sup>                |
| 20     | $E_T$                  | Total spectrum energy                                      | J/m <sup>2</sup>                |
| 21     | $S_\zeta(\omega_w)$    | Spectral Density                                           | m <sup>2</sup> /(rad/s)         |
| 22     | $S_\zeta(\omega_{wn})$ | Wave Spectrum ordinates for each frequency bandwidth       | m <sup>2</sup> /(rad/s)         |
| 23     | $\omega_w$             | Wave frequency                                             | rad/s                           |
| 24     | $\delta\omega$         | Bandwidth of wave frequency                                | rad/s                           |
| 25     | $T_z$                  | Mean number of wave zero up-crossings per second           | s                               |
| 26     | $T_c$                  | Average wave period between two neighboring wave crests    | s                               |
| 27     | $T_z$                  | interval between consecutive wave peaks/crests             | s                               |
| 28     | $J$                    | Peak enhancement function                                  | dimensionless                   |
| 29     | $\gamma$               | Peak enhancement factor                                    | dimensionless                   |
| 30     | $\zeta_a$              | Amplitude                                                  | m                               |
| 31     | $y_i$                  | the dependent variable of the time series being forecasted | m                               |
| 33     | $\phi_i$               | Autoregressive coefficients of AR                          | dimensionless                   |
| 34     | $\beta_k$              | the coefficients of exogenous variables                    | dimensionless                   |
| 35     | $\omega_w$             | Angular Frequency                                          | rad/s                           |
| 36     | $\omega_p$             | Peak Angular Frequency                                     | rad/s                           |

dynamically predicting future wave heights in a user-friendly manner. This finding suggests a need for further investigation, either by employing hybrid models that integrate machine learning and signal decomposition techniques or by using machine learning models separately.

Overall, dynamic forecasts are valuable for long-term planning as they directly serve the goal of ship routing, as their limitations in predicting fluctuations are exclusive to shorter forecast windows. This suggests extending the examined dynamic models into hybrid approaches to enhance predictive accuracy. Therefore, further research is still valid to explore and evaluate whether alternative methodologies and smart models can offer more profound insights and predictions into wave height behavior. Table 15 below outlines the parameter notations, definitions, and corresponding units utilized in this study.

## CRediT authorship contribution statement

**Ashraf Zaghwan:** Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Hossein Enshaeei:** Writing – review & editing, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Hamed Majidiyan:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization. **Yousef Amer:** Writing – review & editing, Visualization, Validation, Supervision, Software, Methodology, Investigation, Formal analysis, Conceptualization.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## References

- Box, G.E.P., Jenkins, G.M., Reinsel, G.C., Ljung, G.M., 2015. *Time Series Analysis: Forecasting and Control*, fifth ed. Wiley.
- Cai, Y., Wen, Y.Q., 2014. Ship route design for avoiding heavy weather and sea conditions. *TransNav (Gdynia, Poland)* 8 (4), 551–556. <https://doi.org/10.12716/1001.08.04.09>.
- Cai, J., Chen, G., Lützen, M., Rytter, N.G.M., 2021. A practical AIS-based route library for voyage planning at the pre-fixtured stage. *Ocean Eng.* 236, 109478. <https://doi.org/10.1016/j.oceaneng.2021.109478>.
- Chen, J., Taylor, P.H., Milne, I.A., Gunawan, D., Zhao, W., 2023. Wave-by-wave prediction for spread seas using a machine learning model with physical understanding. *Ocean Eng.* 285, 115450. <https://doi.org/10.1016/j.oceaneng.2023.115450>.
- Debski, R., Dreżewski, R., 2024. Multi-Objective ship route optimisation using estimation of distribution algorithm. *Appl. Sci.* 14 (13), 5919. <https://doi.org/10.3390/app14135919>.
- Du, W., Li, Y., Zhang, G., Wang, C., Chen, P., Qiao, J., 2021. Estimation of ship routes considering weather and constraints. *Ocean Eng.* 228, 108695. <https://doi.org/10.1016/j.oceaneng.2021.108695>.
- D'Agostini, A., Campos, R.M., Guedes Soares, C., 2024. Assessments of multiple metocean forecasts in the North Atlantic Ocean. *Ocean Eng.* 312, 119199. <https://doi.org/10.1016/j.oceaneng.2024.119199>.
- Furuichi, M., Otsuka, N., 2015. Proposing a common platform of shipping cost analysis of the northern sea route and the Suez Canal route. *Marit. Econ. Logist.* 17 (1), 9–31. <https://doi.org/10.1057/mel.2014.29>.
- Guo, Z., Le, W., Wu, Y., Wang, W., 2019. A multi-step approach framework for freight forecasting of river-sea direct transport without direct historical data. *Sustainability* 11 (5), 4252. <https://doi.org/10.3390/su11154252>.
- Guo, Y., Wang, Y., Zhou, J., Wang, J., 2023. Domain-adapted feature transfer: a generalized framework for short-term vessel speed prediction. *Ocean Eng.* 280, 114536. <https://doi.org/10.1016/j.oceaneng.2023.114536>.
- Harris, I., Sanchez Rodrigues, V., Pettit, S., Beresford, A., Liashko, R., 2018. The impact of alternative routing and packaging scenarios on carbon and sulphate emissions in international wine distribution. *Transportation research. Part D Transp. Environ.* 58, 261–279. <https://doi.org/10.1016/j.trd.2016.08.036>.
- Hildeman, A., Bolin, D., Rychlik, I., 2019. Joint Spatial Modeling of Significant Wave Height and Wave Period Using the SPDE Approach.
- Hildeman, A., Bolin, D., Rychlik, I., 2020. Deformed SPDE models with an application to spatial modeling of significant wave height. <https://doi.org/10.1016/j.spasta.2020.100449>.
- Hildeman, A., Bolin, D., Rychlik, I., 2022. Joint spatial modeling of significant wave height and wave period using the SPDE approach. *Probab. Eng. Mech.* 68, 103203. <https://doi.org/10.1016/j.probabeng.2022.103203>.
- Hoorn, S.V., Knapp, S., 2019. Predicting traffic and risk exposure in the maritime industry. *Saf. Now* 5 (3), 42–50. <https://doi.org/10.3390/safety5030042>.
- Ichsan, M., Pradana, M.F., Noche, B., 2019. Estimation and optimization of the voyage energy efficiency operational indicator (EEOI) on Indonesian sea tollway corridors. *IOP Conf. Ser. Mater. Sci. Eng.* 673 (1), 12024. <https://doi.org/10.1088/1757-899X/673/1/012024>.
- Katimon, A., Shahid, S., Mohsenipour, M., 2018. Modeling water quality and hydrological variables using ARIMA: a case study of Johor river, Malaysia. *Sustain. Water Res. Manag.* 4 (4), 991–998. <https://doi.org/10.1007/s40899-017-0202-8>.
- Khairuddin, N., Aris, A.Z., Elshafie, A., Sheikhy Narany, T., Ishak, M.Y., Isa, N.M., 2019. Efficient forecasting model technique for river stream flow in tropical environment. *Urban Water J.* 16 (3), 183–192. <https://doi.org/10.1080/1573062X.2019.1637906>.
- Kulkarni, K., Li, F., Kondratenko, A.A., Kujala, P., 2024. A voyage-level ship performance modelling approach for the simulation of the Finnish-Swedish winter navigation system. *Ocean Eng.* 295, 116997. <https://doi.org/10.1016/j.oceaneng.2024.116997>.
- Kunicka, M., Litwin, W., 2019. Energy demand of short-range inland ferry with series hybrid propulsion depending on the navigation strategy. *Energies* 12 (18), 3499. <https://doi.org/10.3390/en12183499>.
- Lee, J.S., Cho, I.S., 2022. Extracting the maritime traffic route in Korea based on probabilistic approach using automatic identification system big data. *Appl. Sci.* 12 (2), 635. <https://doi.org/10.3390/app12020635>.
- Lee, J.-S., Lee, H.-T., Cho, I.-S., 2022. Maritime traffic route detection framework based on statistical density analysis from AIS data using a clustering algorithm. *IEEE Access* 10, 23355–23366. <https://doi.org/10.1109/ACCESS.2022.3154363>.
- Liu, D., Rong, H., Soares, C.G., 2023. Shipping route modelling of AIS maritime traffic data at the approach to ports. *Ocean Eng.* 289, 115868. <https://doi.org/10.1016/j.oceaneng.2023.115868>.
- Loake, M.C.A., Astfalck, L.C., Cripps, E.J., 2022. Modelling sea surface wind measurements on Australia's North-West Shelf. *Ocean Eng.* 244, 110308. <https://doi.org/10.1016/j.oceaneng.2021.110308>.
- McCord, M.R., Lee, Y.-K., Lo, H.K., 1999. Ship routing through altimetry-derived ocean currents. *Transp. Sci.* 33 (1), 49. <https://doi.org/10.1287/trsc.33.1.49>.
- Moradi, M.H., Brutsche, M., Wenig, M., Wagner, U., Koch, T., 2022. Marine route optimization using reinforcement learning approach to reduce fuel consumption and consequently minimize CO<sub>2</sub> emissions. *Ocean Eng.* 259, 111882. <https://doi.org/10.1016/j.oceaneng.2022.111882>.
- Mwenda, A., Kuznetsov, D., Mirau, S., 2015. Analyzing the impact of historical data length in non-seasonal ARIMA models forecasting. *Math. Theory Model.* 5, 77–85.
- NDBC - Station 41049 historical data, 2025. National data buoy center. [https://www.ndbc.noaa.gov/station\\_history.php?station=41049](https://www.ndbc.noaa.gov/station_history.php?station=41049).
- Nielsen, U.D., Dietz, J., 2020. Estimation of sea state parameters by the wave buoy analogy with comparisons to third generation spectral wave models. *Ocean Eng.* 216, 107781. <https://doi.org/10.1016/j.oceaneng.2020.107781>.
- Nielsen, U.D., Stredulinsky, D.C., 2012. Sea state estimation from an advancing ship – a comparative study using sea trial data. *Appl. Ocean Res.* 34, 33–44. <https://doi.org/10.1016/j.apor.2011.11.001>.
- Ospina, R., Gondim, J.A.M., Leiva, V., Castro, C., 2023. An overview of forecast analysis with ARIMA models during the COVID-19 pandemic: methodology and case study in Brazil. *Mathematics* 11 (14), 3069. <https://doi.org/10.3390/math11143069>.
- Panigrahi, J.K., Umesh, P.A., 2008. Minimal time ship routing using IRS-P4 (MSMR) analyzed wind fields. *Mar. Geod.* 31 (1), 39–48. <https://doi.org/10.1080/01490410701815157>.
- Panigrahi, J.K., Tripathy, J.K., Umesh, P.A., 2008. Optimum tracking of ship routes in 3g-WAM simulated rough weather using IRS-P4 (MSMR) analysed wind fields. *J. Indian Soc. Remote Sens.* 36 (2), 149–158. <https://doi.org/10.1007/s12524-008-0015-1>.
- Reggiannini, M., Salerno, E., Bacciu, C., D'Errico, A., Lo Duca, A., Marchetti, A., Martinelli, M., Mercurio, C., Mistretta, A., Righi, M., Tampucci, M., Paola, C.D., 2024. Remote sensing for maritime traffic understanding. *Remote Sens.* 16 (3), 557. <https://doi.org/10.3390/rs16030557>.
- Roh, M.-I., 2013a. Determination of an economical shipping route considering the effects of sea state for lower fuel consumption. *Int. J. Nav. Archit. Ocean Eng.* 5 (2), 246–262. <https://doi.org/10.2478/IJNAOE-2013-0130>.
- Roh, M.-I., 2013b. Determination of an economical shipping route considering the effects of sea state for lower fuel consumption. *Int. J. Nav. Archit. Ocean Eng.* 5 (2), 246–262.
- Rong, H., Teixeira, A.P., Guedes Soares, C., 2020. Data mining approach to shipping route characterization and anomaly detection based on AIS data. *Ocean Eng.* 198, 106936. <https://doi.org/10.1016/j.oceaneng.2020.106936>.
- Sahranavard, H., Naseri, M., 2022. Investigating the effect of observational data length on ARIMA modeling and forecasting accuracy: a case study of Korian stream watershed, Iran. *Water science & technology. Water Supply* 22 (11), 8253–8265. <https://doi.org/10.2166/ws.2022.370>.
- Sasmal, K., Miratsu, R., Kodaira, T., Fukui, T., Zhu, T., Waseda, T., 2021. Statistical model representing storm avoidance by merchant ships in the North Atlantic Ocean. *Ocean Eng.* 235, 109163. <https://doi.org/10.1016/j.oceaneng.2021.109163>.

- Sharif, N., Rönnqvist, M., Cordeau, J.-F., Audy, J.-F., Warya, G., Ngo, T., 2024. Multi-objective vessel routing problems with safety considerations: a review. *Maritime Transp. Res.* 7, 100122. <https://doi.org/10.1016/j.martra.2024.100122>.
- Son, W.-J., Lee, H.-T., Cho, I.-S., 2020. A study on the estimation of optimal traffic distribution near breakwater in Busan port. *J. Int. Maritime Safety Environ. Affairs Ship.* 4 (4), 151–161. <https://doi.org/10.1080/25725084.2020.1861824>.
- Sun, W., Tang, S., Liu, X., Zhou, S., Wei, J., 2022. An improved ship weather routing framework for CII reduction accounting for wind-assisted rotors. *J. Mar. Sci. Eng.* 10 (12), 1979. <https://doi.org/10.3390/jmse10121979>.
- Tu, E., Zhang, G., Rachmawati, L., Rajabally, E., Huang, G.-B., 2018. Exploiting AIS data for intelligent Maritime navigation: a comprehensive survey from data to methodology. *IEEE Trans. Intell. Transport. Syst.* 19 (5), 1559–1582. <https://doi.org/10.1109/TITS.2017.2724551>.
- Wang, Z., Fathollahzadeh Attar, N., Khalili, K., Behmanesh, J., Band, S.S., Mosavi, A., Chau, K., 2020. Monthly streamflow prediction using a hybrid stochastic-deterministic approach for parsimonious non-linear time series modeling. *Eng. Appl. Comput. Fluid Mech.* 14 (1), 1351–1372. <https://doi.org/10.1080/19942060.2020.1830858>.
- Weintrit, A., Neumann, T., 2011. *Methods and Algorithms in Navigation : Marine Navigation and Safety of Sea Transportation*, first ed. CRC Press.
- Xu, D., Deng, Y., Xin, P., Zhou, X., 2025. Path planning for large ships in inland waterways considering risk assessment of AIS data. *Ocean Eng.* 342, 122792. <https://doi.org/10.1016/j.oceaneng.2025.122792>.
- Yoon, J.H., Kim, D.H., Yun, S.W., Kim, H.J., Kim, S., 2023. Enhancing container vessel arrival time prediction through past voyage route modeling: a case study of Busan new port. *J. Mar. Sci. Eng.* 11 (6), 1234. <https://doi.org/10.3390/jmse11061234>.
- Zhang, Y., Hu, H., Dai, L., 2020. Real-time assessment and prediction on maritime risk state on the Arctic route. *Marit. Pol. Manag.* 47 (3), 352–370. <https://doi.org/10.1080/03088839.2019.1693064>.